

ARTICLE KD-3 FATIGUE EVALUATION

KD-300 SCOPE

This Article presents a traditional fatigue analysis design approach. In accordance with KD-140, if it can be shown that the vessel will fail in a leak-before-burst mode, then the number of design cycles shall be calculated in accordance with either Article KD-3 or Article KD-4. When performing analysis using methods of Article KD-3 on vessels where leak-before-burst has been established (see KD-141), welded vessel construction details shall be analyzed using the Structural Stress method (see KD-340). The Structural Stress method shall only be used for welded details. If a leak-before-burst mode of failure cannot be shown, then the number of design cycles shall be calculated in accordance with Article KD-4.

KD-301 GENERAL

Cyclic operation may cause fatigue failure of pressure vessels and components. While cracks often initiate at the bore, cracks may initiate at outside surfaces or at layer interfaces for autofrettaged and layered vessels. In all cases, areas of stress concentrations are a particular concern. Fatigue-sensitive points shall be identified and a fatigue analysis made for each point. The result of the fatigue analysis will be a calculated number of design cycles N_f for each type of operating cycle, and a calculated cumulative effect number of design cycles when more than one type of operating cycle exists.

The resistance to fatigue of a nonwelded component shall be based on the design fatigue curves for the materials used. Fatigue resistance of weld details shall be determined using the Structural Stress method (see KD-340), which is based on fatigue data of actual welds.

In some cases it may be convenient or necessary to obtain experimental fatigue data for a nonwelded component itself rather than for small specimens of the material (see KD-1260). If there are two or more types of stress cycles which produce significant stresses, their cumulative effect shall be evaluated by calculating for each type of stress cycle the usage factors U_1, U_2, U_3 , etc., and the cumulative usage factor U per KD-330. The cumulative usage factor U shall not exceed 1.0.

KD-302 THEORY

The theory used in this Article postulates that fatigue at any point is controlled by the alternating stress intensity S_{alt} and the associated mean stress σ_{nm} normal to the plane of S_{alt} . They are combined to define the equivalent alternating stress intensity S_{eq} , which is used with the design fatigue curves to establish the number of design cycles N_f .

KD-302.1 Alternating Stress Intensity. The alternating stress intensity S_{alt} represents the maximum range of shear stress.

KD-302.2 Associated Mean Stress. The associated mean stress σ_{nm} is the mean value of stress normal to the plane subjected to the maximum alternating stress intensity.

For welded construction, austenitic stainless steel construction, and high-strength low alloy steel bolting, the associated mean stress shall not be combined with the alternating stress intensity [see KD-312.4(a)].

KD-310 STRESS ANALYSIS FOR FATIGUE EVALUATION

The calculation of the number of design cycles shall be based on a stress analysis of all fatigue-sensitive points.

KD-311 LOADING CONDITIONS AND RESIDUAL STRESSES

In this analysis, consideration shall be taken of the following loadings and stresses.

KD-311.1 Residual Stresses Due to Manufacturing.

(a) Some manufacturing processes such as forming, etc., introduce residual tensile stresses of unknown magnitude. Unless these stresses are controlled by some method, such as postfabrication heat treatment or mechanical overstrain processes like autofrettage, these initial residual stresses shall be assumed to have a peak magnitude corresponding to the yield strength of the material.

(b) Manufacturing processes such as welding, heat treatment, forming, autofrettage, shrink fitting, and wire wrapping introduce residual stresses. Tensile residual stresses shall be included in the calculation of associated mean stresses. Compressive residual stresses may also be included. When calculating the residual stresses introduced by autofrettage, due account shall be taken of the influence of the Bauschinger effect (see Article KD-5). If any combination of operational or hydrotest loadings will produce yielding at any point, any resulting change in the residual stress values shall be taken into account.

(c) In welded construction, no credit shall be taken for beneficial residual stresses within the weld metal or the heat-affected zone.

(d) In austenitic stainless steel construction, no credit shall be taken for beneficial residual stresses.

KD-311.2 Operating Stresses. Mean and alternating stresses shall be calculated for all loading conditions specified in the User's Design Specification. Stress concentration factors shall be determined by analytical or experimental techniques.

Ranges of stress intensities due to cyclic loadings and associated mean stresses (residual plus operational) shall be calculated on the assumption of elastic behavior. If these calculations show that yielding occurs, a correction shall be made. see KD-312.3.

KD-312 CALCULATION OF FATIGUE STRESSES WHEN PRINCIPAL STRESS DIRECTIONS DO NOT CHANGE

For any case in which the directions of the principal stresses at the point being considered do not change during the operating cycle, the methods stated in KD-312.1 through KD-312.4 shall be used to determine the fatigue controlling stress components.

KD-312.1 Principal Stresses. Determine the values of the three principal stresses at the point being investigated for the complete operating cycle assuming the loading and conditions described in KD-311. These stresses are designated σ_1 , σ_2 , and σ_3 .

- (13) **KD-312.2 Alternating Stress Intensities.** Determine the stress differences (maintain the proper algebraic sign for the complete operating cycle):

$$S_{12} = \sigma_1 - \sigma_2 \quad (\text{KD-312.1})$$

$$S_{23} = \sigma_2 - \sigma_3 \quad (\text{KD-312.2})$$

$$S_{31} = \sigma_3 - \sigma_1 \quad (\text{KD-312.3})$$

In the following, the symbol S_{ij} is used to represent any one of these three differences.

Identify the algebraic largest stress difference $S_{ij \max}$ and the algebraic smallest difference $S_{ij \min}$ of each S_{ij} during the complete operating cycle. Then the alternating stress intensity $S_{alt \ ij}$ is determined by:

$$S_{alt \ ij} = 0.5(S_{ij \max} - S_{ij \min}) \quad (\text{KD-312.4})$$

The absolute magnitude of these three alternating stress intensities ($S_{alt \ 12}$, $S_{alt \ 23}$, and $S_{alt \ 31}$) are the three ranges of shear stress that shall be considered in a fatigue analysis. Each will have an associated mean stress (determined below), which also influences the fatigue behavior.

- (13) **KD-312.3 Associated Mean Stress.**

(a) For nonwelded construction using the design fatigue curve of Figure KD-320.2, austenitic stainless steel construction, and high-strength low alloy steel bolting, see KD-312.4(a).

(b) For nonwelded construction, the associated mean stresses $\sigma_{nm \ ij}$ shall be calculated in accordance with the following method.

The stresses σ_n normal to the plane of the maximum shear stress, associated with the three $S_{alt \ ij}$, are given by:

$$\sigma_{n \ 12} = 0.5(\sigma_1 + \sigma_2) \quad (\text{KD-312.5})$$

$$\sigma_{n \ 23} = 0.5(\sigma_2 + \sigma_3) \quad (\text{KD-312.6})$$

$$\sigma_{n31} = 0.5(\sigma_3 + \sigma_1) \quad (\text{KD-312.7})$$

In the following, the symbol $\sigma_{n\ ij}$ is used to represent any one of these normal stresses.

Identify the maximum $\sigma_{n\ ij\ \max}$ and the minimum $\sigma_{n\ ij\ \min}$ value of each $\sigma_{n\ ij}$ during the complete operating cycle. Then the mean normal stresses $\sigma_{nm\ ij}$ shall be calculated by:

For autofrettaged vessels or nonautofrettaged vessel with compressive mean stress

(1) when $S_{ij\ \max} < S_y$ and $S_{ij\ \min} > -S_y$, then

$$\sigma_{nm\ ij} = 0.5(\sigma_{n\ ij\ \max} + \sigma_{n\ ij\ \min}) \quad (\text{KD-312.8})$$

(2) when $S_{alt\ ij} \geq S_y$, then

$$\sigma_{nm\ ij} = 0 \quad (\text{KD-312.9})$$

If neither (1) nor (2) applies, then the stress values used in this analysis shall be determined from an elastic-plastic analysis with ideally elastic-plastic (nonstrain-hardening) material using S_y . S_y is yield strength at operating temperature from Table Y-1 of Section II, Part D. Alternatively, $\sigma_{nm\ ij}$ may be calculated as equal to $0.5(\sigma_{n\ ij\ \max} + \sigma_{n\ ij\ \min})$ but not less than zero.

For nonautofrettaged vessels with tensile mean stress

(3) when $S_{ij\ \max} < S_y/2$ and $S_{ij\ \min} > -S_y/2$, then

$$\sigma_{nm\ ij} = 0.5(\sigma_{n\ ij\ \max} + \sigma_{n\ ij\ \min}) \quad (\text{KD-312.10})$$

(4) when $S_{alt\ ij} \geq S_y/2$, then

$$\sigma_{nm\ ij} = 0 \quad (\text{KD-312.11})$$

If neither (3) nor (4) applies, then the stress values used in this analysis shall be determined from an elastic-plastic analysis with ideally elastic (nonstrain hardening) material using $S_y/2$. Alternatively, $\sigma_{nm\ ij}$ may be calculated as equal to $0.5(\sigma_{n\ ij\ \max} + \sigma_{n\ ij\ \min})$ but not less than 0.

(13) **KD-312.4 Equivalent Alternating Stress Intensity.**

(a) For nonwelded construction made of carbon or low alloy steels having an ultimate tensile strength less than 90 ksi (620 MPa) (see Figure KD-320.2), and for austenitic stainless steel and nonwelded aluminum alloy construction (see Figure KD-320.3), effects of mean stresses are incorporated in the design fatigue curve. Therefore:

$$S_{eq\ ij} = S_{alt\ ij} \quad (\text{KD-312.12})$$

(b) For nonwelded construction, the equivalent alternating stress intensity S_{eq} , which is assumed to have the same effect on fatigue as the combination of the alternating stress intensity S_{alt} and its associated mean stress σ_{nm} , shall be calculated in accordance with the equation:

$$S_{eq\ ij} = S_{alt\ ij} \frac{1}{1 - \beta \sigma_{nm\ ij} / S'_a} \quad (\text{KD-312.13})$$

where S'_a is the allowable amplitude of the alternating stress component when $\sigma_{nm} = 0$ and $N = 10^6$ cycles (see KD-321). The value of β shall be 0.2 for carbon or low alloy steel forged nonwelded construction (see Figure KD-320.1). The value of β shall be 0.2 for $\sigma_{nm\ ij} < 0$ and 0.5 for $\sigma_{nm\ ij} > 0$ for 17-4PH or 15-5PH stainless steel nonwelded construction using forgings or bar (see Figure KD-320.4). Other values of β may be used if justified by experimental evidence. If the values of $\beta \sigma_{nm\ ij} / S'_a$ exceeds 0.9, limit its value to 0.9.

Using this equation, three values of $S_{eq\ ij}$ are obtained. The largest of these three shall be used in combination with the design fatigue curve to establish the number of design cycles in accordance with KD-322(a) or KD-322(f).

KD-313 CALCULATION OF FATIGUE STRESSES WHEN PRINCIPAL STRESS AXES CHANGE

When the directions of the principal stresses change during the loading cycle, the plane carrying the maximum range of shear stress cannot be easily identified using equations based on principal stresses. The position of each plane at the point of interest can be defined by two angles and a convenient set of Cartesian axes. By varying this combination of angles in increments, it is possible to determine the range of shear stress on each plane. The largest of these shear stress ranges shall be considered to be the alternating stress intensity, S_{alt} , used in the calculation of design cycles.

KD-320 CALCULATED NUMBER OF DESIGN CYCLES

The calculation of the number of design cycles N_f shall be based either on design fatigue curves described in KD-321 or on results of experimental fatigue tests on components as stated in KD-1260.

(13) KD-321 BASIS FOR DESIGN FATIGUE CURVES

(a) The conditions and procedures of this paragraph are based on a comparison between the calculated equivalent alternating stress intensity S_{eq} and strain cycling fatigue data. The strain cycling fatigue data have been used to derive design fatigue curves. These curves show the allowable amplitude S_a of the alternating stress component (one-half of the alternating stress range) plotted against the number of design cycles N_f , which the component is assumed to safely endure without failure.

(b) The design fatigue curves have been derived from strain-controlled push-pull tests with zero mean stress (i.e., $\sigma_{nm} = 0$) on polished unnotched specimens in dry air. The imposed strains have been multiplied by the elastic modulus and a design margin has been provided so as to make the calculated equivalent stress intensity amplitude and the allowable stress amplitude directly comparable. S_{eq} and S_a have the dimensions of stress, but they do not represent a real stress when the elastic range is exceeded.

(c) The design fatigue curves for forged nonwelded construction presented in Figure KD-320.1 have been developed from fatigue tests in dry air with polished specimens of steels having an ultimate tensile strength in the range of 90 ksi to 180 ksi (620 MPa to 1 200 MPa). Fatigue tests with small cylinders pressurized from the inside by oil and made of low alloy steels having an ultimate tensile strength in the range of 130 ksi to 180 ksi (900 MPa to 1 200 MPa) have been used to confirm the validity of these curves for carbon or low alloy forgings with machined surfaces. For design fatigue curves, see Figure KD-320.1 (and Table KD-320.1) for forged carbon or low alloy steel construction, Figure KD-320.2 for nonwelded carbon and low alloy steels having an ultimate tensile strength less than 90 ksi (620 MPa), Figure KD-320.3 for austenitic stainless steel construction, Figure KD-320.4 for 17-4PH or 15-5PH stainless steel construction, and Figure KD-320.5 for high-strength low alloy steel bolting.

(d) The design fatigue curves are not applicable in the presence of aggressive environments. For conditions not covered by these design fatigue curves, the Manufacturer shall provide supplementary fatigue data.

(13) KD-322 USE OF DESIGN FATIGUE CURVE

(a) Figure KD-320.1 shall be used for forged nonwelded parts with machined surfaces made of carbon or low alloy steels having a specified minimum value of the ultimate tensile strength S_u greater than 90 ksi (620 MPa). The curves are applicable for an average surface roughness of 19 R_a microinch (0.5 micrometers) or a maximum surface roughness of 59 microinch (1.5 micrometers) R_{max} (peak-to-valley height) in fatigue-sensitive areas. Lower quality surface finish will influence fatigue. This influence is considered by a factor K_r [see Figure KD-320.6(a) or Figure KD-320.6(b)], which shall be combined with S_{eq} as specified in (g) when determining the calculated number of design cycles N_f .

(b) Figure KD-320.2 shall be used for nonwelded parts with machined surfaces made of carbon and low alloy steels having an ultimate tensile strength less than 90 ksi (620 MPa). The influence of the surface roughness is included in the curve, i.e., $K_r = 1.0$; therefore, a surface roughness factor need not be applied.

(c) Figure KD-320.3 shall be used for forged nonwelded parts with machined surfaces made of austenitic stainless steels. The influence of the surface roughness is included in the curve, i.e., $K_r = 1.0$; therefore, a surface roughness factor need not be applied.

(d) Figure KD-320.4 shall be used for forged nonwelded parts with machined surfaces made of 17-4PH/15-5PH stainless steel having ultimate tensile strength S_u of 115 ksi (793 MPa) or greater. The curve is applicable for an average surface roughness of 19 R_a microinch (0.5 micrometer) or a maximum surface roughness of 59 microinch (1.5 micrometer) R_{max} in fatigue-sensitive areas. Lower quality surface finish will influence fatigue. This influence is considered by a factor K_r [see Figure KD-320.6(a) or Figure KD-320.6(b)], which shall be combined with S_{eq} as specified in (f) when determining the calculated number of design cycles N_f .

(e) Figure KD-320.7 or Figure KD-320.7M shall be used for nonwelded 6061-T6 and 6061-T651 aluminum alloys.

(f) High-strength alloy steel bolts and studs may be evaluated for cyclic operation by the methods of KD-3 using the design fatigue curve of Figure KD-320.5, provided

(1) the material is one of the following:

(-a) SA-193, Grade B7; SA-193, B16; SA-320, L7

(-b) SA-320, L7M and SA-320, L43

(2) "V" type threads shall have a minimum thread root radius no smaller than 0.032 times the pitch, and in no case smaller than 0.004 in. (0.102 mm).

(3) fillet radii at the end of the shank shall be such that the ratio of the fillet radius to shank diameter is not less than 0.060.

The bolt stress shall be determined using the root area. See KD-627 for the fatigue strength reduction factor for threads. K_r may be assumed to be 1.0 when Figure KD-320.5 is used. The designer should use caution in calculating bolt load from applied torque. The designer shall consider that corrosion effects on a bolted connection can reduce bolt fatigue life.

(g) When the operational cycle being considered is the only one that produces significant fluctuating stresses, the calculated number of design cycles N_f is determined as follows.

(1) Evaluate the fatigue penalty factor, K_e , using ΔS_n and the following equations where parameters m and n are determined from Table KD-322.1:

$$K_e = 1.0 \text{ for } \Delta S_n \leq 2S_y \quad (\text{KD-322.1})$$

$$K_e = 1.0 + \frac{(1-n)}{n(m-1)} \left(\frac{\Delta S_n}{2S_y} - 1 \right) \text{ for } 2S_y < \Delta S_n < 2mS_y \quad (\text{KD-322.2})$$

$$K_e = \frac{1}{n} \text{ for } \Delta S_n \geq 2mS_y \quad (\text{KD-322.3})$$

where ΔS_n is the primary-plus-secondary stress intensity range.

Otherwise, the alternate method given in KD-323 may be used to calculate K_e .

(2) Identify the applicable fatigue curve for the material as explained in (a).

(3) Enter the curve from the ordinate axis at the value:

$$S_a = K_r K_e S_{eq} \frac{E(\text{curve})}{E(\text{analysis})} \quad (\text{KD-322.4})$$

where

$E(\text{curve})$ = Modulus of Elasticity given on the Design Fatigue Curve

$E(\text{analysis})$ = Modulus of Elasticity used in the analysis

(4) Read the corresponding number of cycles on the abscissa. This is the calculated number of design cycles N_f .

(13) **KD-323 ALTERNATIVE METHOD FOR EVALUATING THE FATIGUE PENALTY FACTOR, K_e**

The fatigue penalty factor, K_e , may be calculated using the following operations:

$$K_e = \frac{(\Delta \epsilon_t)_{ep}}{(\Delta \epsilon_t)_e} \quad (\text{KD-323.1})$$

where

$$(\Delta \epsilon_t)_{ep} = \frac{\sqrt{2}}{3} [(\Delta e_{11} - \Delta e_{22})^2 + (\Delta e_{22} - \Delta e_{33})^2 + (\Delta e_{33} - \Delta e_{11})^2 + 1.5(\Delta e_{12}^2 + \Delta e_{23}^2 + \Delta e_{31}^2)]^{0.5} \quad (\text{KD-323.2})$$

$$(\Delta \epsilon_t)_e = \frac{\Delta \sigma_e}{E_{ya}} \quad (\text{KD-323.3})$$

$$\Delta\sigma_e = \frac{1}{\sqrt{2}} [(\Delta\sigma_1 - \Delta\sigma_2)^2 + (\Delta\sigma_2 - \Delta\sigma_3)^2 + (\Delta\sigma_3 - \Delta\sigma_1)^2]^{0.5} \quad (\text{KD-323.4})$$

and where

- E_{ya} = the modulus of elasticity at the point under consideration, evaluated at the mean temperature of the cycle
 $(\Delta\epsilon_t)_{ep}$ = the equivalent total strain range from the elastic-plastic analysis for the points of interest
 $(\Delta\epsilon_t)_e$ = the equivalent total strain range from the elastic analysis for the points of interest
 $\Delta\sigma_e$ = the range of primary-plus-secondary-plus-peak equivalent stress

(13) **KD-330 CALCULATED CUMULATIVE EFFECT NUMBER OF DESIGN CYCLES**

If there are two or more types of stress cycles which produce significant stresses, the alternating stress intensity and the associated mean stress shall be calculated for each type of stress cycle. The cumulative effect of all of the stress cycles shall be evaluated using a linear damage relationship as specified in (a) through (f).

(a) Calculate the number of times each type of stress cycle of type 1, 2, 3, etc., will be repeated during a specific design service life period L . It is recommended that L be based on the design service L_d as specified in the User's Design Specification; designate these numbers n_1, n_2, n_3 , etc., or generally n_i .

(b) For each type of stress cycle, determine S_a by the procedures given in KD-312.4. Designate these quantities S_{a1}, S_{a2}, S_{a3} , etc., or generally S_{ai} .

(c) For each value S_{ai} , use the applicable design fatigue curve to determine the maximum number of design repetitions N_i if this type of cycle were the only one acting. Designate these as N_1, N_2, N_3 , etc., or generally N_i .

(d) For each type of stress cycle, calculate the usage factor $U_i = n_i/N_i$.

(e) Calculate the cumulative usage factor from:

$$U = \sum_{i=1}^j \frac{n_i}{N_i}, \text{ or } U_1 + U_2 \dots \quad (\text{KD-330.1})$$

The cumulative usage factor U shall not exceed 1.0.

(f) Calculate the design service L_d using the equation:

$$L_d = L / U \quad (\text{KD-330.2})$$

KD-340 FATIGUE ASSESSMENT OF WELDS — ELASTIC ANALYSIS AND STRUCTURAL STRESS

(a) An equivalent structural stress range parameter is used to evaluate the fatigue damage for results obtained from a linear elastic stress analysis. The controlling stress for the fatigue evaluation is the structural stress that is a function of the membrane and bending stresses normal to the hypothetical crack plane.

(b) Fatigue cracks at pressure vessel welds are typically located at the toe of a weld. For as-welded and weld joints subject to post weld heat treatment, the expected orientation of a fatigue crack is along the weld toe in the through-thickness direction, and the structural stress normal to the expected crack is the stress measure used to correlate fatigue life data. For fillet welded components, fatigue cracking may occur at the toe of the fillet weld or the weld throat, and both locations shall be considered in the assessment. It is difficult to accurately predict fatigue life at the weld throat due to variability in throat dimension, which is a function of the depth of the weld penetration. It is recommended to perform sensitivity analysis where the weld throat dimension is varied.

(13) **KD-341 ASSESSMENT PROCEDURE**

The following procedure can be used to evaluate protection against failure due to cyclic loading using the equivalent structural stress range.

Step 1. Determine a load history based on the information in the User's Design Specification and the histogram development methods in KD-350. The load history should include all significant operating loads and events that are applied to the component.

Step 2. For a location at a weld joint subject to a fatigue evaluation, determine the individual stress-strain cycles using the cycle counting methods in KD-350. Define the total number of cyclic stress ranges in the histogram as M .

2013 SECTION VIII - DIVISION 3

Step 3. Determine the elastically calculated membrane and bending stress normal to the assumed hypothetical crack plane at the start and end points (time points ${}^m t$ and ${}^n t$, respectively) for the k^{th} cycle counted in Step 2. See Nonmandatory Appendix L for guidance on linearization of stress results to obtain membrane and bending stresses. Using this data, calculate the membrane and bending stress ranges between time points ${}^m t$ and ${}^n t$, and the maximum, minimum, and mean stress.

$$\Delta\sigma_{m,k}^e = {}^m\sigma_{m,k}^e - {}^n\sigma_{m,k}^e \quad (\text{KD-341.1})$$

$$\Delta\sigma_{b,k}^e = {}^m\sigma_{b,k}^e - {}^n\sigma_{b,k}^e \quad (\text{KD-341.2})$$

$$\sigma_{max,k} = \max\left[({}^m\sigma_{m,k}^e + {}^m\sigma_{b,k}^e), ({}^n\sigma_{m,k}^e + {}^n\sigma_{b,k}^e)\right] \quad (\text{KD-341.3})$$

$$\sigma_{min,k} = \min\left[({}^m\sigma_{m,k}^e + {}^m\sigma_{b,k}^e), ({}^n\sigma_{m,k}^e + {}^n\sigma_{b,k}^e)\right] \quad (\text{KD-341.4})$$

$$\sigma_{mean,k} = \frac{\sigma_{max,k} + \sigma_{min,k}}{2} \quad (\text{KD-341.5})$$

Step 4. Determine the elastically calculated structural stress range for the k^{th} cycle, $\Delta\sigma_k^e$, using eq. (KD-341.6).

$$\Delta\sigma_k^e = \Delta\sigma_{m,k}^e + \Delta\sigma_{b,k}^e \quad (\text{KD-341.6})$$

Step 5. Determine the elastically calculated structural strain, $\Delta\varepsilon_k^e$, from the elastically calculated structural stress, $\Delta\sigma_k^e$, using eq. (KD-341.7).

$$\Delta\varepsilon_k^e = \frac{\Delta\sigma_k^e}{E_{ya,k}} \quad (\text{KD-341.7})$$

The corresponding local nonlinear structural stress and strain ranges, $\Delta\sigma_k$ and $\Delta\varepsilon_k$, respectively, are determined by simultaneously solving Neuber's Rule, eq. (KD-341.8), and a model for the material hysteresis loop stress-strain curve given by eq. (KD-341.9); see KD-360.

$$\Delta\sigma_k \cdot \Delta\varepsilon_k = \Delta\sigma_k^e \cdot \Delta\varepsilon_k^e \quad (\text{KD-341.8})$$

$$\Delta\varepsilon_k = \frac{\Delta\sigma_k}{E_{ya,k}} + 2\left(\frac{\Delta\sigma_k}{2K_{css}}\right)^{\frac{1}{n_{css}}} \quad (\text{KD-341.9})$$

The structural stress range computed solving eqs. (KD-341.8) and (KD-341.9) is subsequently modified for low-cycle fatigue using eq. (KD-341.10).

$$\Delta\sigma_k = \left(\frac{E_{ya,k}}{1 - \nu^2}\right)\Delta\varepsilon_k \quad (\text{KD-341.10})$$

NOTE: The modification for low-cycle fatigue should always be performed because the exact distinction between high-cycle fatigue and low-cycle fatigue cannot be determined without evaluating the effects of plasticity which is a function of the applied stress range and cyclic stress-strain curve. For high-cycle fatigue applications, this procedure will provide correct results, i.e., the elastically calculated structural stress will not be modified.

Step 6. Compute the equivalent structural stress range parameter for the k^{th} cycle using the following equations. In eq. (KD-341.11), for SI units, the thickness, t , stress range, $\Delta\sigma_k$, and the equivalent structural stress range parameter, $\Delta S_{ess, k}$, are in mm, MPa, and MPa/(mm) $^{(2-m_{ss})/2m_{ss}}$, respectively, and for U.S. Customary units, the thickness, t , stress range, $\Delta\sigma_k$, and the equivalent structural stress range parameter, $\Delta S_{ess, k}$, are in in., ksi, and ksi/(in.) $^{(2-m_{ss})/2m_{ss}}$, respectively.

$$\Delta S_{ess, k} = \frac{\Delta\sigma_k}{t_{ess}^{\left(\frac{2-m_{ss}}{2m_{ss}}\right)} \cdot I_{m_{ss}} \cdot f_{M, k}} \quad (KD-341.11)$$

where

$$m_{ss} = 3.6 \quad (KD-341.12)$$

$$t_{ess} = 16 \text{ mm } (0.625 \text{ in.}) \text{ for } t \leq 16 \text{ mm } (0.625 \text{ in.}) \quad (KD-341.13)$$

$$t_{ess} = t \text{ for } 16 \text{ mm } (0.625 \text{ in.}) < t < 150 \text{ mm } (6 \text{ in.}) \quad (KD-341.14)$$

$$t_{ess} = 150 \text{ mm } (6 \text{ in.}) \text{ for } t \geq 150 \text{ mm } (6 \text{ in.}) \quad (KD-341.15)$$

$$I_{m_{ss}} = \frac{1.23 - 0.364R_{b, k} - 0.17R_{b, k}^2}{1.007 - 0.306R_{b, k} - 0.178R_{b, k}^2} \quad (KD-341.16)$$

$$R_{b, k} = \frac{|\Delta\sigma_{b, k}^e|}{|\Delta\sigma_{m, k}^e| + |\Delta\sigma_{b, k}^e|} \quad (KD-341.17)$$

$$f_{M, k} = (1 - R_k)^{\frac{1}{m_{ss}}} \text{ for } \begin{cases} \sigma_{mean, k} \geq 0.5S_{y, k} \text{ and} \\ R_k > 0, \text{ and} \\ |\Delta\sigma_{m, k}^e + \Delta\sigma_{b, k}^e| \leq 2S_{y, k} \end{cases} \quad (KD-341.18)$$

$$f_{M, k} = 1.0 \text{ for } \begin{cases} \sigma_{mean, k} < 0.5S_{y, k} \text{ or} \\ R_k \leq 0, \text{ or} \\ |\Delta\sigma_{m, k}^e + \Delta\sigma_{b, k}^e| > 2S_{y, k} \end{cases} \quad (KD-341.19)$$

$$R_k = \frac{\sigma_{min, k}}{\sigma_{max, k}} \quad (KD-341.20)$$

Step 7. Determine the permissible number of cycles, N_k , based on the equivalent structural stress range parameter for the k^{th} cycle computed in Step 6. Fatigue curves for welded joints are provided in KD-370.

Step 8. Determine the fatigue damage for the k^{th} cycle, where the actual number of repetitions of the k^{th} cycle is n_k .

$$D_{f, k} = \frac{n_k}{N_k} \quad (KD-341.21)$$

Step 9. Repeat Steps 6 through 8 for all stress ranges, M , identified in the cycle counting process in Step 3.

Step 10. Compute the accumulated fatigue damage using the following equation. The location along the weld joint is suitable for continued operation if this equation is satisfied.

$$D_f = \sum_{i=1}^M D_{f,k} \leq 1.0 \quad (\text{KD-341.22})$$

Step 11. Repeat Steps 5 through 10 for each point along the weld joint that is subject to a fatigue evaluation.

(13) KD-342 ASSESSMENT PROCEDURE MODIFICATIONS

The assessment procedure in KD-341 may be modified as shown below.

(a) *Multiaxial Fatigue*. If the structural shear stress range is not negligible, i.e., $\Delta\tau_k > \Delta\sigma_k/3$, a modification should be made when computing the equivalent structural stress range. Two conditions need to be considered:

(1) If $\Delta\sigma_k$ and $\Delta\tau_k$ are out of phase, the equivalent structural stress range, $\Delta S_{ess, k}$, in eq. Step 6, eq. (KD-341.11) should be replaced by

$$\Delta S_{ess, k} = \frac{1}{F(\delta)} \left[\left(\frac{\Delta\sigma_k}{t_{ess} \left(\frac{2 - m_{ss}}{2m_{ss}} \right) \cdot I_{m,ss}^{-1} \cdot f_{M,k}} \right)^2 + 3 \left(\frac{\Delta\tau_k}{t_{ess} \left(\frac{2 - m_{ss}}{2m_{ss}} \right) \cdot I_{\tau,ss}^{-1}} \right)^2 \right]^{0.5} \quad (\text{KD-342.1})$$

where

$$I_{\tau,ss}^{-1} = \frac{1.23 - 0.364R_{b\tau,k} - 0.17R_{b\tau,k}^2}{1.007 - 0.306R_{b\tau,k} - 0.178R_{b\tau,k}^2} \quad (\text{KD-342.2})$$

$$R_{b\tau,k} = \frac{|\Delta\tau_{b,k}^e|}{|\Delta\tau_{m,k}^e| + |\Delta\tau_{b,k}^e|} \quad (\text{KD-342.3})$$

$$\Delta\tau_k = \Delta\tau_{m,k}^e + \Delta\tau_{b,k}^e \quad (\text{KD-342.4})$$

$$\Delta\tau_{m,k}^e = m_{\tau_{m,k}}^e - n_{\tau_{m,k}}^e \quad (\text{KD-342.5})$$

$$\Delta\tau_{b,k}^e = m_{\tau_{b,k}}^e - n_{\tau_{b,k}}^e \quad (\text{KD-342.6})$$

In eq. (KD-342.1), $F(\delta)$ is a function of the out-of-phase angle between $\Delta\sigma_k$ and $\Delta\tau_k$ if both loading modes can be described by sinusoidal functions, or:

$$F(\delta) = \frac{1}{\sqrt{2}} \left(1 + \left[1 - \frac{12 \cdot \Delta\sigma_k^2 \cdot \Delta\tau_k^2 \cdot \sin^2(\delta)}{(\Delta\sigma_k^2 + 3\Delta\tau_k^2)^2} \right]^{0.5} \right)^{0.5} \quad (\text{KD-342.7})$$

A conservative approach is to ignore the out-of-phase angle and recognize the existence of a minimum possible value for $F(\delta)$ in eq. (KD-342.7) given by:

$$F(\delta) = \frac{1}{\sqrt{2}} \quad (\text{KD-342.8})$$

(2) If $\Delta\sigma_k$ and $\Delta\tau_k$ are in-phase, the equivalent structural stress range, $\Delta S_{ess, k}$, is given by eq. (KD-342.1) with $F(\delta) = 1.0$.

(b) *Weld Quality*. If a defect exists at the toe of a weld that can be characterized as a crack-like flaw, i.e., undercut, and this defect exceeds the value permitted by ASME Section VIII, Division 2, Part 7, then a reduction in fatigue life shall be calculated by substituting the value of $I^{1/m}$ in eqs. (KD-341.16) and (KD-342.2) with the value given by eq. (KD-342.9). In this equation, a is the depth of the crack-like flaw at the weld toe. Equation (KD-342.9) is valid only when $a/t \leq 0.1$.

$$\frac{1}{I^{m_{ss}}} = \frac{1.229 - 0.365R_{b,k} + 0.789\left(\frac{a}{t}\right) - 0.17R_{b,k}^2 + 13.771\left(\frac{a}{t}\right)^2 + 1.243R_{b,k}\left(\frac{a}{t}\right)}{1 - 0.302R_{b,k} + 7.115\left(\frac{a}{t}\right) - 0.178R_{b,k}^2 + 12.903\left(\frac{a}{t}\right)^2 - 4.091R_{b,k}\left(\frac{a}{t}\right)} \quad (\text{KD-342.9})$$

KD-350 HISTOGRAM DEVELOPMENT AND CYCLE COUNTING FOR FATIGUE ANALYSIS

This paragraph contains cycle counting procedures required to perform a fatigue assessment for irregular stress or strain versus time histories. These procedures are used to break the loading history down into individual cycles that can be evaluated using the fatigue assessment rules of KD-340. Two cycle counting methods are presented in this paragraph. An alternative cycle counting method may be used if agreed to by the Owner-User.

KD-351 DEFINITIONS

The definitions used in this section are shown below.

cycle: a relationship between stress and strain that is established by the specified loading at a location in a vessel or component. More than one stress-strain cycle may be produced at a location, either within an event or in transition between two events, and the accumulated fatigue damage of the stress-strain cycles determines the adequacy for the specified operation at that location. This determination shall be made with respect to the stabilized stress-strain cycle.

event: the User's Design Specification may include one or more events that produce fatigue damage. Each event consists of loading components specified at a number of time points over a time period and is repeated a specified number of times. For example, an event may be the startup, shutdown, upset condition, or any other cyclic action. The sequence of multiple events may be specified or random.

nonproportional loading: if the orientation of the principal axes are not fixed, but changes orientation during cyclic loading, the loading is called nonproportional. An example of nonproportional loading is a shaft subjected to out-of-phase torsion and bending, where the ratio of axial and torsional stress varies continuously during cycling.

peak: the point at which the first derivative of the loading or stress histogram changes from positive to negative.

proportional loading: during constant amplitude loading, as the magnitudes of the applied stresses vary with time, the size of Mohr's circle of stress also varies with time. In some cases, even though the size of Mohr's circle varies during cyclic loading, if the orientation of the principal axes remains fixed, the loading is called proportional. An example of proportional loading is a shaft subjected to in-phase torsion and bending, where the ratio of axial and torsional stress remains constant during cycling.

valley: the point at which the first derivative of the loading or stress histogram changes from negative to positive.

KD-352 HISTOGRAM DEVELOPMENT

The loading histogram should be determined based on the specified loadings provided in the User's Design Specification. The loading histogram should include all significant operating loads and events that are applied to the component. The following should be considered in developing the loading histogram:

- (a) The number of repetitions of each event during the operation life.
- (b) The sequence of events during the operation life, if applicable.
- (c) Applicable loadings such as pressure, temperature, supplemental loads such as weight, support displacements, and nozzle reaction loadings.
- (d) The relationship between the applied loadings during the time history.

(13) KD-353 CYCLE COUNTING USING THE RAINFLOW METHOD

The Rainflow Cycle Counting Method ASTM E1049 is recommended to determine the time points representing individual cycles for the case of situations where the variation in time of loading, stress, or strain can be represented by a single parameter. This cycle counting method is not applicable for nonproportional loading. Cycles counted with the Rainflow Method correspond to closed stress-strain hysteresis loops, with each loop representing a cycle.

KD-353.1 Recommended Procedure.

Step 1. Determine the sequence of peaks and valleys in the loading histogram. If multiple loadings are applied, it may be necessary to determine the sequence of peaks and valleys using a stress histogram. If the sequence of events is unknown, the worst case sequence should be chosen.

Step 2. Reorder the loading histogram to start and end at either the highest peak or lowest valley, so that only full cycles are counted. Determine the sequence of peaks and valleys in the loading history. Let X denote the range under consideration, and let Y denote the previous range adjacent to X .

Step 3. Read the next peak or valley. If out of data, go to Step 8.

Step 4. If there are less than three points, go to Step 3; if not, form ranges X and Y using the three most recent peaks and valleys that have not been discarded.

Step 5. Compare the absolute values of ranges X and Y .

(a) If $X < Y$, go to Step 3

(b) If $X \geq Y$, go to Step 6

Step 6. Count range Y as one cycle; discard the peak and valley of Y . Record the time points and loadings or component stresses, as applicable, at the starting and ending time points of the cycle.

Step 7. Return to Step 4 and repeat Steps 4 through 6 until no more time points with stress reversals remain.

Step 8. Using the data recorded for the counted cycles, perform fatigue assessment in accordance with this Article.

KD-354 CYCLE COUNTING USING MAX-MIN CYCLE COUNTING METHOD

The Max-Min Cycle Counting Method is recommended to determine the time points representing individual cycles for the case of nonproportional loading. The cycle counting is performed by first constructing the largest possible cycle, using the highest peak and lowest valley, followed by the second largest cycle, etc., until all peak counts are used.

(13) KD-354.1 Recommended Procedure.

Step 1. Determine the sequence of peaks and valleys in the loading history. If some events are known to follow each other, group them together, but otherwise arrange the random events in any order.

Step 2. Calculate the elastic stress components, σ_{ij} , produced by the applied loading at every point in time during each event at a selected location of a vessel. All stress components must be referred to the same global coordinate system. The stress analysis must include peak stresses at local discontinuities.

Step 3. Scan the interior points of each event and delete the time points at which none of the stress components indicate reversals (peaks or valleys).

Step 4. Using the stress histogram from Step 2, determine the time point with the highest peak or lowest valley. Designate the time point as ${}^m t$, and the stress components as ${}^m \sigma_{ij}$.

Step 5. If time point ${}^m t$ is a peak in the stress histogram, determine the component stress range between time point ${}^m t$ and the next valley in the stress histogram. If time point ${}^m t$ is a valley, determine the component stress range between time point ${}^m t$ and the next peak. Designate the next time point as ${}^n t$, and the stress components as ${}^n \sigma_{ij}$. Calculate the stress component ranges and the von Mises equivalent stress range between time points ${}^m t$ and ${}^n t$.

$${}^{mn} \Delta \sigma_{ij} = {}^m \sigma_{ij} - {}^n \sigma_{ij} \quad (\text{KD-354.1})$$

$$\begin{aligned} & {}^{mn} \Delta S_{\text{range}} \\ &= \frac{1}{\sqrt{2}} \left[\left({}^{mn} \Delta \sigma_{11} - {}^{mn} \Delta \sigma_{22} \right)^2 + \left({}^{mn} \Delta \sigma_{22} - {}^{mn} \Delta \sigma_{33} \right)^2 + \left({}^{mn} \Delta \sigma_{33} - {}^{mn} \Delta \sigma_{11} \right)^2 + 6 \left({}^{mn} \Delta \sigma_{12}^2 + {}^{mn} \Delta \sigma_{23}^2 + {}^{mn} \Delta \sigma_{31}^2 \right) \right]^{0.5} \end{aligned} \quad (\text{KD-354.2})$$

Step 6. Repeat Step 5 for the current point, ${}^m t$, and the time point of the next peak or valley in the sequence of the stress histogram. Repeat this process for every remaining time point in the stress histogram.

Step 7. Determine the maximum von Mises equivalent stress range obtained in Step 5 and record the time points ${}^m t$ and ${}^n t$ that define the start and end points of the k^{th} cycle.

Step 8. Determine the event or events to which the time points ${}^m t$ and ${}^n t$ belong and record their specified number of repetitions as ${}^m N$ and ${}^n N$, respectively.

Step 9. Determine the number of repetitions of the k^{th} cycle.

(a) If ${}^mN < {}^nN$, delete the time point mt from those considered in Step 4 and reduce the number of repetitions at time point nt from nN to $({}^nN - {}^mN)$.

(b) If ${}^mN > {}^nN$, delete the time point nt from those considered in Step 4 and reduce the number of repetitions at time point mt from mN to $({}^mN - {}^nN)$.

(c) If ${}^mN = {}^nN$, delete both time points mt and nt from those considered in Step 4.

Step 10. Return to Step 4 and repeat Steps 4 through 10 until no more time points with stress reversals remain.

Step 11. Using the data recorded for the counted cycles, perform fatigue assessment in accordance with this Article.

(13) KD-360 CYCLIC STRESS-STRAIN CURVE

The cyclic stress-strain curve of a material (i.e., strain amplitude versus stress amplitude) may be represented by eq. (KD-360.1). The material constants for this model are provided in Table KD-360.1.

$$\epsilon_{ta} = \frac{\sigma_a}{E_y} + \left[\frac{\sigma_a}{K_{css}} \right]^{\frac{1}{n_{css}}} \quad (\text{KD-360.1})$$

The hysteresis loop stress-strain curve of a material (i.e., strain range versus stress range) obtained by scaling the cyclic stress-strain curve by a factor of two is represented by eq. (KD-360.2). The material constants provided in Table KD-360.1 are also used in this equation.

$$\epsilon_{tr} = \frac{\sigma_r}{E_y} + 2 \left[\frac{\sigma_r}{2K_{css}} \right]^{\frac{1}{n_{css}}} \quad (\text{KD-360.2})$$

KD-370 WELDED JOINT DESIGN FATIGUE CURVES

KD-371 FATIGUE CURVE MATERIALS

Subject to the limitations of KD-340, eq. (KD-372.1) in KD-372(a) can be used to evaluate design fatigue life of welded joints for the following materials and associated temperature limits:

(a) Carbon, Low Alloy, Series 4xx, and High Tensile Strength Steels for temperatures not exceeding 700°F (371°C)

(b) Series 3xx High Alloy Steels, Nickel-Chromium-Iron Alloy, Nickel-Iron-Chromium Alloy, and Nickel-Copper Alloy for temperatures not exceeding 800°F (427°C)

(c) Wrought 70 Copper-Nickel for temperatures not exceeding 450°F (232°C)

(d) Nickel-Chromium-Molybdenum-Iron, Alloys X, G, C-4, and C-276 for temperatures not exceeding 800°F (427°C)

(e) Aluminum Alloys for temperatures not exceeding 225°F (107°C)

(13) KD-372 DESIGN CYCLE COMPUTATION

The number of allowable design cycles for the welded joint fatigue curve shall be computed as follows:

(a) The design number of allowable design cycles, N , can be computed from eq. (KD-372.1) based on the equivalent structural stress range parameter, $\Delta S_{ess,k}$, determined in accordance with KD-340 of this Division. The constants C and h for use in eq. (KD-372.1) are provided in Table KD-370.1. The lower 99% Prediction Interval (-3σ) shall be used for design unless otherwise agreed to by the Owner-User and the Manufacturer.

$$N = \frac{f_I}{f_E} \left(\frac{f_{MT} \cdot C}{\Delta S_{ess,k}} \right)^{\frac{1}{h}} \quad (\text{KD-372.1})$$

(b) If a fatigue improvement method is performed that exceeds the fabrication requirements of this Division, then a fatigue improvement factor, f_i , may be applied. The fatigue improvement factors shown below may be used. An alternative factor determined may also be used if agreed to by the user or user's designated agent and the Manufacturer.

2013 SECTION VIII - DIVISION 3

(1) For burr grinding in accordance with Figure KD-372.1

$$f_I = 1.0 + 2.5 \cdot (10)^q \quad (\text{KD-372.2})$$

(2) For TIG dressing

$$f_I = 1.0 + 2.5 \cdot (10)^q \quad (\text{KD-372.3})$$

(3) For hammer peening

$$f_I = 1.0 + 4.0 \cdot (10)^q \quad (\text{KD-372.4})$$

In the above equations, the parameter q is given by the following equation:

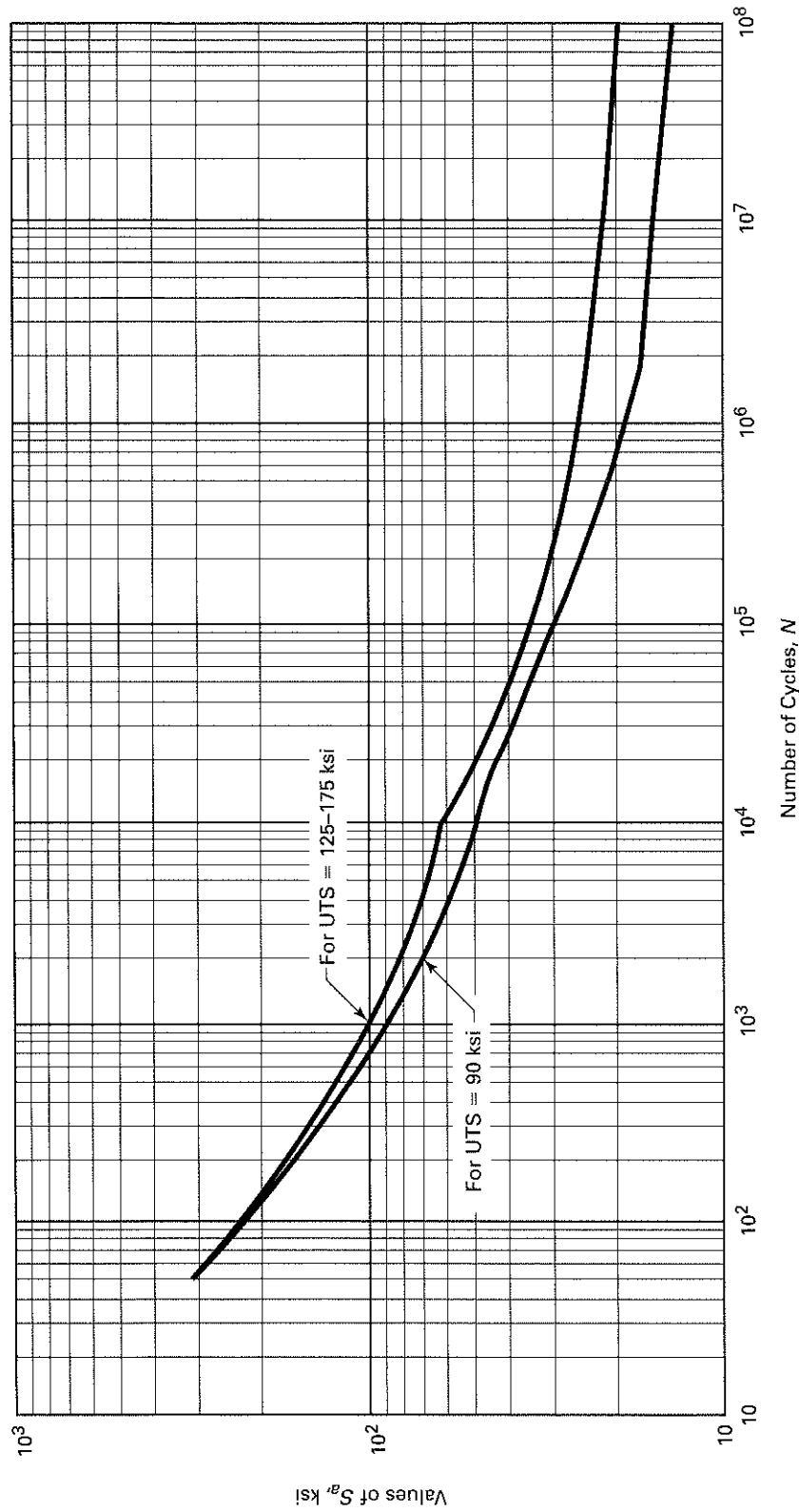
$$q = -0.0016 \cdot \left(\frac{\Delta S_{ess,k}}{C_{us}} \right)^{1.6} \quad (\text{KD-372.5})$$

(c) The design fatigue cycles given by eq. (KD-372.1) may be modified to account for the effects of environment other than dry ambient air that may cause corrosion or subcritical crack propagation. The environmental modification factor, f_E , is typically a function of the fluid environment, loading frequency, temperature, and material variables such as grain size and chemical composition. A value of $f_E = 4.0$ shall be used unless there is specific information to justify an alternate value based on the severity of the material/environmental interaction. A value of $f_E = 1.0$ may be used for dry ambient air. The environmental modification factor, f_E , shall be specified in the User's Design Specification.

(d) A temperature adjustment is required to the fatigue curve for materials other than carbon steel and/or for temperatures above 21°C (70°F). The temperature adjustment factor is given by eq. (KD-372.6).

$$f_{MT} = \frac{E_T}{E_{ACS}} \quad (\text{KD-372.6})$$

Figure KD-320.1
Design Fatigue Curves $S_a = f(N_f)$ for Nonwelded Machined Parts Made of Forged Carbon or Low Alloy Steels for Temperatures Not Exceeding 700°F



GENERAL NOTES:

- (a) $E = 28.3 \times 10^6$ psi
- (b) Interpolate for UTS = 90 - 125 ksi
- (c) Table KD-320.1 contains tabulated values and equations for an accurate interpolation of these curves.

Table KD-320.1
Tabulated Values of S_a , ksi, From Figures Indicated

Figure	Curve	Number of Design Operating Cycles, N_f																			
		5E1	1E2	2E2	5E2	1E3	2E3	5E3	1E4	2E4	5E4	1E5	2E5	5E5	1E6	2E6	5E6	1E7	2E7	5E7	1E8
320.1	UTS 90 ksi	311	226	164	113	89	72	57	49	43	34	29	25	21	19	17	16.2	15.7	15.2	14.5	14
320.1	UTS 125 ksi	317	233	171	121	98	82	68	61	49	39	34	31	28	26	24	22.9	22.1	21.4	20.4	19.7
320.2	UTS ≤ 80 ksi	275	205	155	105	83	64	48	38	31	23	20	16.5	13.5	12.5	12.1	11.5	11.1	10.8	10.3	9.9
320.2	UTS 115-130 ksi	230	175	135	100	78	62	49	44	36	29	26	24	22	20	19.3	18.5	17.8	17.2	16.4	15.9
320.3	Austenitic stainless steel	345	261	201	148	119	97	76	64	56	46	41	36	31	28	...	...	...	...	...	...
320.4	17-4PH/ 15-5PH stainless steel	205	171	149	129	103	86.1	72.0	65.1	60.0	54.8	51.6	48.7	45.2	42.8	40.6	37.8	35.9	...	...	...
320.5	HSLA steel bolting	450	300	205	122	81	55	33	22.5	15	10.5	8.4	7.1	6.0	5.3	...	...	...	...	...	...

GENERAL NOTES:

- (a) All notes on the referenced figures apply to these data.
 (b) Number of design cycles indicated shall be read as follows: 1E/ = 1×10^1 , e.g., 5E2 = 5×10^2 or 500 cycles.
 (c) Interpolation between tabular values is permissible based upon data representation by straight lines on a log-log plot. Accordingly, for $S_i > S > S_j$,

$$\frac{N}{N_i} = \left(\frac{N_j}{N_i} \right)^{\left[\log(S_i/S) / \log(S_i/S_j) \right]}$$

where

S, S_i, S_j = values of S_a

N, N_i, N_j = corresponding calculated number of design cycles from design fatigue data

For example, from the data above, use the interpolation equation above to find the calculated number of design cycles N for $S_a = 50.0$ ksi when UTS ≥ 125 ksi on Figure KD-320.1:

$$\frac{N}{10,000} = \left(\frac{20,000}{10,000} \right)^{\left[\log(61/50) / \log(61/49) \right]}$$

$N = 18,800 \text{ cycles}$

(d) Equations for number of design operating cycles:

(1) Figure KD-320.1, UTS = 90 ksi

$$311 \text{ ksi} \geq S_a \geq 42.6 \text{ ksi} \quad N = \text{EXP} \left[15.433 - 2.0301 \ln(S_a) + 1036.035 \ln(S_a / S_a^2) \right]$$

$$17 \text{ ksi} < S_a < 42.6 \text{ ksi} \quad N = \left[(2.127\text{E-}05) + (7.259\text{E-}10) S_a^3 - (8.636\text{E-}06) \ln(S_a) \right]^{-1}$$

$$17 \text{ ksi} \geq S_a \geq 14 \text{ ksi} \quad N = \text{EXP}[-20.0 \ln(S_a / 35.12)]$$

(2) Figure KD-320.1, UTS = 125-175 ksi

$$317 \text{ ksi} \geq S_a \geq 60.6 \text{ ksi} \quad N = \left[0.00122 - (7.852\text{E-}05) S_a + (7.703\text{E-}06) S_a^{1.5} \right]^{-1}$$

$$24 \text{ ksi} < S_a < 60.6 \text{ ksi} \quad N = \left[\left[(7.8628\text{E-}05) + (3.212\text{E-}03) S_a + (9.36\text{E-}02) S_a^2 \right] / \left[1 - (8.599\text{E-}02) S_a + (1.816\text{E-}03) S_a^2 + (4.05774\text{E-}06) S_a^3 \right] \right]^2$$

$$24 \text{ ksi} \geq S_a \geq 19.7 \text{ ksi} \quad N = \text{EXP}[-20.0 \ln(S_a / 49.58)]$$

Table KD-320.1
Tabulated Values of S_a , ksi, From Figures Indicated (Cont'd)

GENERAL NOTES (CONT'D):

(3) Figure KD-320.2, UTS $\leq$ 80 ksi $S_a \geq 38$ ksi

$$N = \left[- (7.125E-04) + (4.4692E-08)(S_a^2) \ln(S_a) + (3.561E-03) / S_a^{0.5} \right]^{-1}$$

12.5 ksi $< S_a < 38$ ksi

$$N = \text{EXP} \left[\left[18.0353 - 1.3663S_a - (1.549E-03)S_a^2 \right] / \left[1 - (4.031E-02)S_a - (3.854E-03)S_a^2 \right] \right]$$

 $S_a \leq 12.5$ ksi

$$N = \text{EXP} \left[-20.0 \ln(S_a / 24.94) \right]$$

(4) Figure KD-320.2, UTS = 115-130 ksi

 $S_{alt} \geq 43$ ksi

$$N = \text{EXP} \left[\left[9.363 - 3.004E-01(S_{alt}) + 1.488E-04(S_{alt}^2) \right] / \left[1 - 2.4133E-02(S_{alt}) - 1.6829E-04(S_{alt}^2) \right] \right]$$

20 ksi $< S_{alt} < 43$ ksi

$$N = \left(-1974.51 + 1063.7(S_{alt}^{0.5}) - 146.64(S_{alt}) / (1 - 6.73933E-01(S_{alt}^{0.5}) + 1.51483E-01(S_{alt}) - 1.1358E-02(S_{alt}^{1.5})) \right)$$

 $S_{alt} \leq 20$ ksi

$$N = \text{EXP} \left[-20.0 \ln(S_{alt} / 39.91) \right]$$

(5) Figure KD-320.3, austenitic stainless steel

345 ksi $\geq S_a \geq 55.7$ ksi

$$N = \text{EXP} \left[\left[(3.03E-02) - 0.7531S_a - (1.968E-04)S_a^2 \right] / \left[1 - (7.23E-02)S_{eq} - (4.075E-04)S_a^2 \right] \right]$$

55.7 ksi $> S_a \geq 28$ ksi

$$N = \text{EXP} \left[\left[(2.445E-04) + (1.656E-03)S_a - (3.416E-02)S_a^2 \right] / \left[1 - (6.062E-02)S_a - (4.29E-04)S_a^2 - (4.049E-05)S_a^3 \right] \right]$$

(6) Figure KD-320.4, 17-4PH/15-5PH stainless steel

129 ksi $\leq S_a < 207$ ksi

$$N = 10^{(-10.600[\log(S_a)]^3 + 80.024[\log(S_a)]^2 - 203.37[\log(S_a)] + 175.13)}$$

103 ksi $\leq S_a < 129$ ksi

$$N = 10^{(-10.600[\log(S_a)]^3 + 80.024[\log(S_a)]^2 - 203.37[\log(S_a)] + 175.13)}$$

71 ksi $\leq S_a < 103$ ksi

$$N = 10^{(-29.577[\log(S_a)]^3 + 180.59[\log(S_a)]^2 - 370.92[\log(S_a)] + 259.15)}$$

71 ksi $> S_a \geq 35.7$ ksi

$$N = 10^{(-29.577[\log(S_a)]^3 + 180.59[\log(S_a)]^2 - 370.92[\log(S_a)] + 259.15)}$$

(7) Figure KD-320.5, HSLA steel bolting

450 ksi $\geq S_a > 81$ ksi

$$N = -5.1633E1 + (2.3417)S_a^{0.5} - (2.0909E4) / S_a + (9.3946E5) / S_a^{1.5}$$

81 ksi $\geq S_a > 22.5$ ksi

$$N = -3.5678E1 + (4.0293E5)S_a^{1.5} + (3.1693E6) / S_a^2$$

22.5 ksi $\geq S_a > 8.4$ ksi

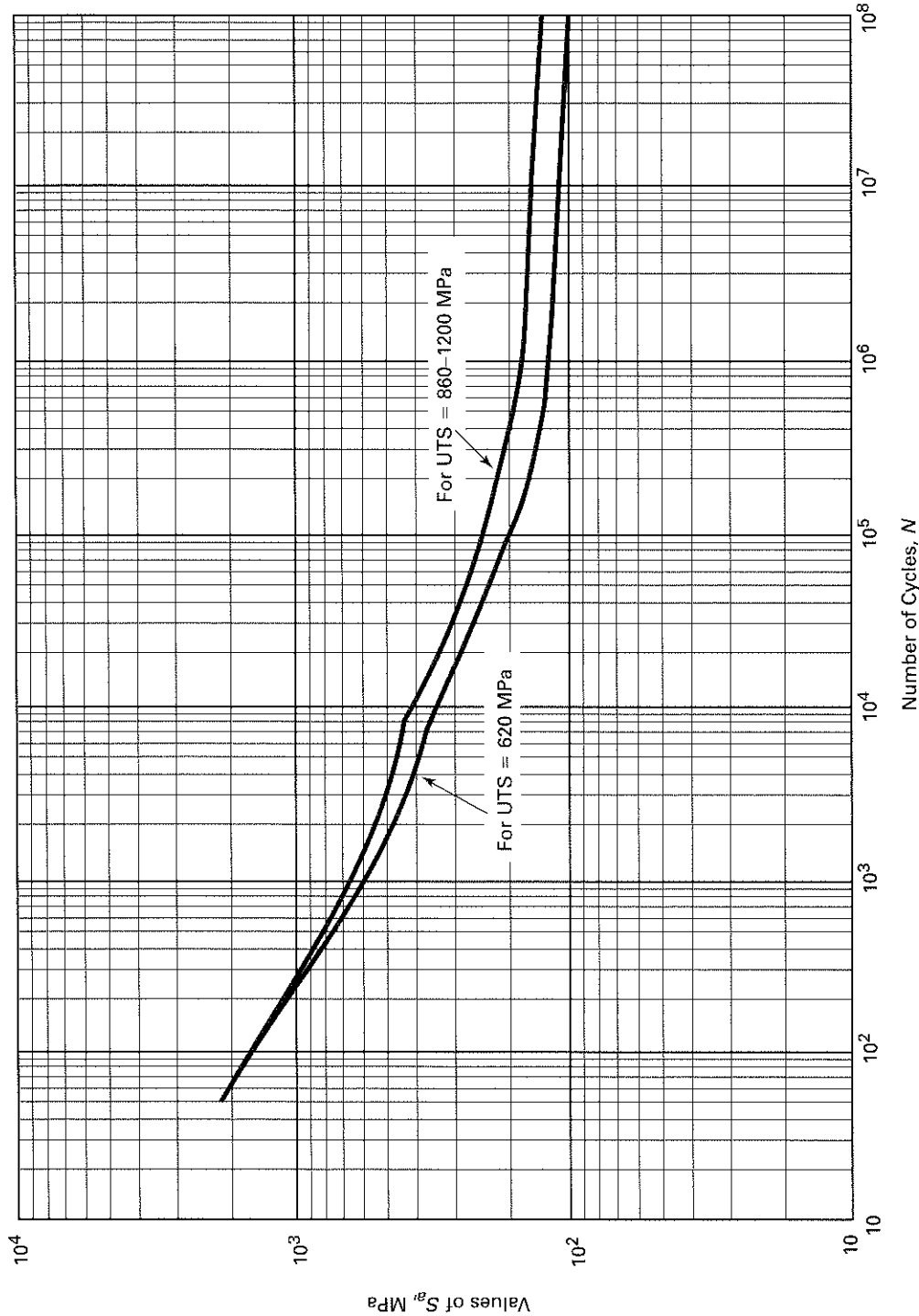
$$N = \text{EXP} \left[1.2990 + (3.2307E-1) \ln(S_a)^2 + (3.4538E1) \ln(S_a) / S_a \right]$$

8.4 ksi $\geq S_a \geq 5.3$ ksi

$$N = \text{EXP} \left[1.8243E1 - (3.9800E-1)S_a^2 + (1.0422 \cdot E1)S_a^{2.5} \right]$$

(e) Equations shall not be used outside of the cycle range given in the Table.

Figure KD-320.1M
Design Fatigue Curves $S_a = f(N_f)$ for Nonwelded Machined Parts Made of Forged Carbon or Low Alloy Steels for Temperatures Not Exceeding 371°C



GENERAL NOTES:

- (a) $E = 195 \times 10^3$ MPa
- (b) Interpolate for UTS = 620 - 860 MPa
- (c) Table KD-320.1M contains tabulated values and equations for these curves, and a equation for an accurate interpolation of these curves.

(13)

Table KD-320.1M
Tabulated Values of S_a , MPa, From Figures Indicated

Figure	Curve	Number of Design Operating Cycles, N_f																			
		5E1	1E2	2E2	5E2	1E3	2E3	5E3	1E4	2E4	5E4	1E5	2E5	5E5	1E6	2E6	5E6	1E7	2E7	5E7	1E8
320.1M	UTS 620 MPa	2140	1560	1130	779	614	496	393	338	297	234	200	172	145	131	117.2	111.8	108.2	104.8	100	96.5
320.1M	UTS 860 MPa	2190	1610	1200	834	675	565	469	421	338	269	234	214	193	179	165	157.9	152.4	147.6	140.7	135.8
320.2M	UTS ≤ 552 MPa	1900	1410	1070	724	572	441	331	262	214	159	138	114	93	86	83	79	77	74	71	68
320.2M	UTS 793 – 896 MPa	1586	1207	931	690	538	427	338	303	248	200	179	165	152	138	133	127	123	119	113	110
320.3M	Austenitic stainless steel	2380	1800	1390	1020	821	669	524	441	386	317	283	248	214	193	...	...	...	...	...	...
320.4M	17-4PH/15-5PH stainless steel	1410	1180	1030	889	710	594	496	449	414	378	356	336	312	295	280	261	248	...	...	...
320.5M	HSLA steel bolting	3100	2070	1410	841	558	379	228	155	103	72	58	49	41	37	...	...	...	...	...	...

GENERAL NOTES:

(a) All notes on the referenced figures apply to these data.

(b) Number of design cycles indicated shall be read as follows: 1EJ = 1×10^J , e.g., 5E2 = 5×10^2 or 500 cycles.(c) Interpolation between tabular values is permissible based upon data representation by straight lines on a log-log plot. Accordingly, for $S_i > S > S_j$,

$$\frac{N}{N_i} = \left(\frac{N_j}{N_i} \right)^{\left[\frac{\log(S_i/S)}{\log(S_i/S_j)} \right]}$$

where

 S, S_i, S_j = values of S_a N, N_i, N_j = corresponding calculated number of design cycles from design fatigue dataFor example, from the data above, use the interpolation equation above to find the calculated number of design cycles N for $S_a = 50.0$ ksi when UTS ≥ 125 ksi on Figure KD-320.1:

$$\frac{N}{10,000} = \left(\frac{20,000}{10,000} \right)^{\left[\frac{\log(61/50)}{\log(61/49)} \right]}$$

$$N = 18,800 \text{ cycles}$$

(d) Equations for number of design operating cycles:

(1) Figure KD-320.1M, UTS = 620 MPa

$$2140 \text{ MPa} \geq S_a \geq 294 \text{ MPa} \quad N = \text{EXP}[19.353 - 2.0301 \ln(S_a) + \{49254.16 \ln(S_a) - 95099.7\} / S_a^2]$$

$$117 \text{ MPa} < S_a < 294 \text{ MPa} \quad N = [(3.794\text{E-}05) + \{2.297\text{E-}12\} S_a^3 - \{8.636\text{E-}06\} \ln(S_a)]^{-1}$$

$$117 \text{ MPa} \geq S_a \geq 96.5 \text{ MPa} \quad N = \text{EXP}[-20.0 \ln(S_a / 242.11)]$$

(2) Figure KD-320.1M, UTS = 860–1210 MPa

$$2190 \text{ MPa} \geq S_a \geq 418 \text{ MPa} \quad N = [0.00122 - \{1.139\text{E-}05\} S_a + \{4.255\text{E-}07\} S_a^{1.5}]^{-1}$$

$$166 \text{ MPa} < S_a < 418 \text{ MPa} \quad N = \left[\frac{[(7.8628\text{E-}05) + \{4.659\text{E-}04\} S_a + \{1.97\text{E-}03\} S_a^2] / \{1 - (1.247\text{E-}02) S_a + \{3.820\text{E-}05\} S_a^2 + (1.238\text{E-}08) S_a^3\}}{2} \right]^{-2}$$

$$166 \text{ MPa} \geq S_a \geq 135.8 \text{ MPa} \quad N = \text{EXP}[-20.0 \ln(S_a / 341.81)]$$

(3) Figure KD-320.2M, UTS ≤ 552 MPa

$$S_a \geq 262 \text{ MPa} \quad N = \left[-\{7.125\text{E-}04\} + \{9.401\text{E-}10\} (S_a^2) \ln(S_a) - \{1.8512\text{E-}09\} S_a^2 + \{9.35\text{E-}03\} / S_a^{0.5} \right]^{-1}$$

$$86 \text{ MPa} < S_a < 262 \text{ MPa} \quad N = \text{EXP}[(18.0353 - 0.19617 S_a - \{3.258\text{E-}04\} S_a^2) / (1 - \{5.846\text{E-}03\} S_a - \{8.107\text{E-}05\} S_a^2)]$$

$$S_a \leq 86 \text{ MPa} \quad N = \text{EXP}[-20.0 \ln(S_a / 171.96)]$$

Table KD-320.1M
Tabulated Values of S_a , MPa, From Figures Indicated (Cont'd)

GENERAL NOTES (CONT'D):

(4) Figure KD-320.2M, UTS = 793–896 MPa

$$S_{alt} \geq 296 \text{ MPa} \quad N = \text{EXP} \left[\left(9.2095 - 4.3897\text{E-}02(S_{alt}) + 3.3186\text{E-}06(S_{alt}^2) \right) / \left(1 - 3.60\text{E-}03(S_{alt}) - 3.4936\text{E-}06(S_{alt}^2) \right) \right]$$

$$138 \text{ MPa} < S_{alt} < 296 \text{ MPa} \quad N = \left(-1974.51 + 405.043(S_{alt}^{0.5}) - 21.2626(S_{alt}) / \left(1.0 - 0.256625(S_{alt}^{0.5}) + 2.19648\text{E-}02(S_{alt}) - 6.27115\text{E-}04(S_{alt}^{1.5}) \right) \right)$$

$$S_{alt} \leq 138 \text{ MPa} \quad N = \text{EXP}[-20.0 \ln(S_{alt} / 275.35)]$$

(5) Figure KD-320.3M, austenitic stainless steel

$$2380 \text{ MPa} \geq S_a \geq 384 \text{ MPa} \quad N = \text{EXP} \left[\left(0.0303 - 0.1092S_a - (4.140\text{E-}06)S_a^2 \right) / \left(1 - (1.05\text{E-}02)S_a - (8.572\text{E-}06)S_a^2 \right) \right]$$

$$384 \text{ MPa} > S_a \geq 193 \text{ MPa} \quad N = \text{EXP} \left[\left((2.445\text{E-}04) + (2.402\text{E-}04)S_a - (7.186\text{E-}04)S_a^2 \right) / \left(1 - (8.789\text{E-}03)S_a - (9.02\text{E-}06)S_a^2 - (1.235\text{E-}07)S_a^3 \right) \right]$$

(6) Figure KD-320.4M, 17-4PH/15-5PH stainless steel

$$889 \text{ MPa} \leq S_a < 1410 \text{ MPa} \quad N = 10^{(-10.600[\log(S_a)]^3 + 106.689[\log(S_a)]^2 - 359.932[\log(S_a)] + 408.175)}$$

$$710 \text{ MPa} \leq S_a < 889 \text{ MPa} \quad N = 10^{(56.735[\log(S_a)]^3 - 490.48[\log(S_a)]^2 + 1410.54[\log(S_a)] - 1346.471)}$$

$$490 \text{ MPa} \leq S_a < 710 \text{ MPa} \quad N = 10^{(-29.577[\log(S_a)]^3 + 254.993[\log(S_a)]^2 - 736.164[\log(S_a)] + 714.587)}$$

$$490 \text{ MPa} > S_a \geq 248 \text{ MPa} \quad N = 10^{(41.740[\log(S_a)]^3 - 306.509[\log(S_a)]^2 + 737.234[\log(S_a)] - 573.962)}$$

(7) Figure KD-320.5M, HSLA steel bolting

$$3100 \text{ MPa} \geq S_a > 558 \text{ MPa} \quad N = -7.3187\text{E}1 + (2.8331\text{E}4) / S_a^{0.5} - (6.5611\text{E}5) \ln(S_a) / S_a + (4.0791\text{E}6) / S_a$$

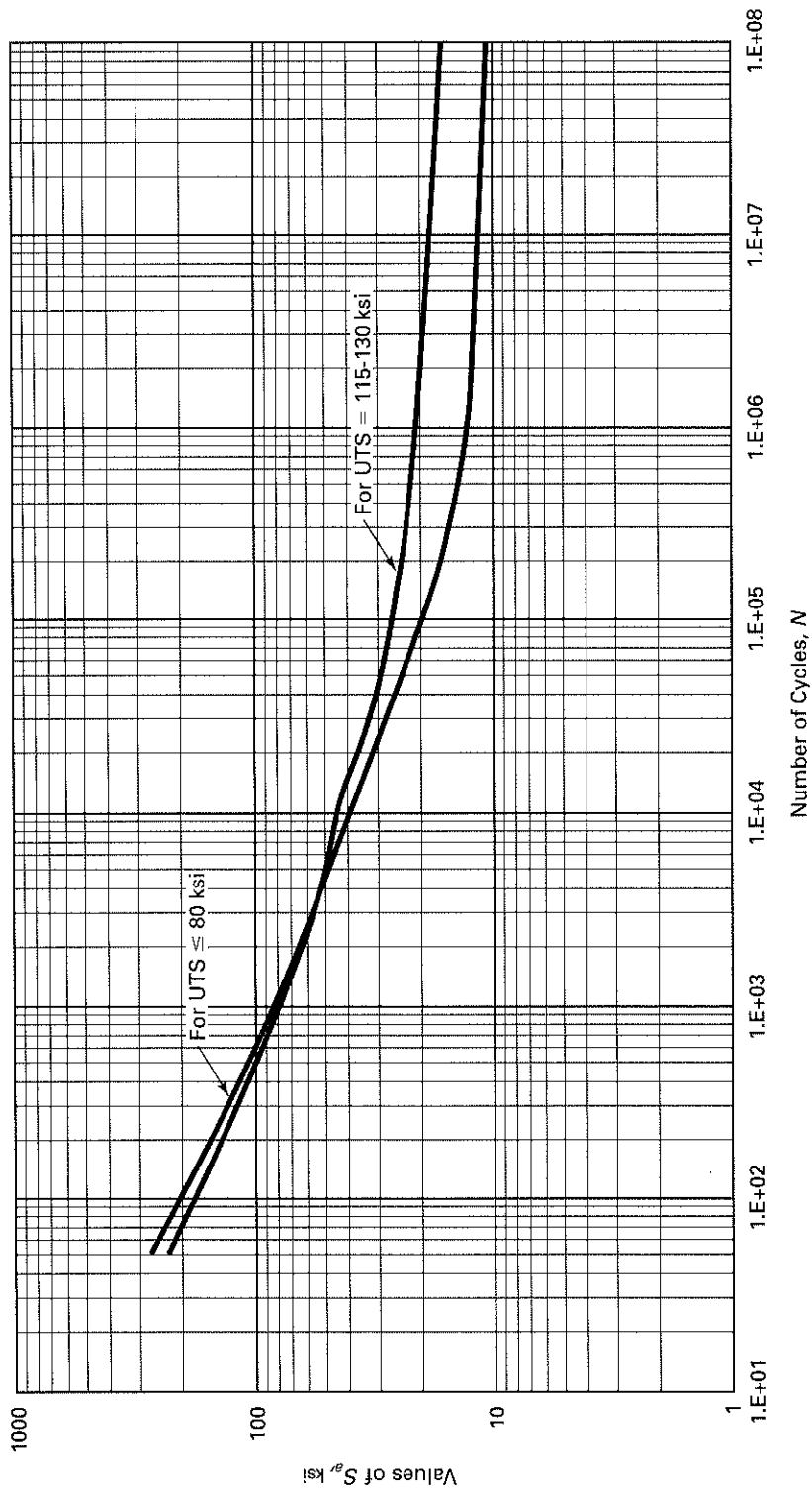
$$558 \text{ MPa} \geq S_a > 155 \text{ MPa} \quad N = 1.9878\text{E}1 + (1.1798\text{E}4) / S_a + (4.7180\text{E}7) \ln(S_a) / S_a^2$$

$$155 \text{ MPa} \geq S_a > 58 \text{ MPa} \quad N = \text{EXP}[4.4080\text{E}1 + (1.1258) \ln(S_a)^2 - (1.2592\text{E}1) \ln(S_a)]$$

$$58 \text{ MPa} \geq S_a \geq 37 \text{ MPa} \quad N = [7.6208\text{E-}6 - (5.6599\text{E-}7)S_a + (1.0461\text{E-}8)S_a^2]^{-1}$$

(e) Equations shall not be used outside of the cycle range given in the Table.

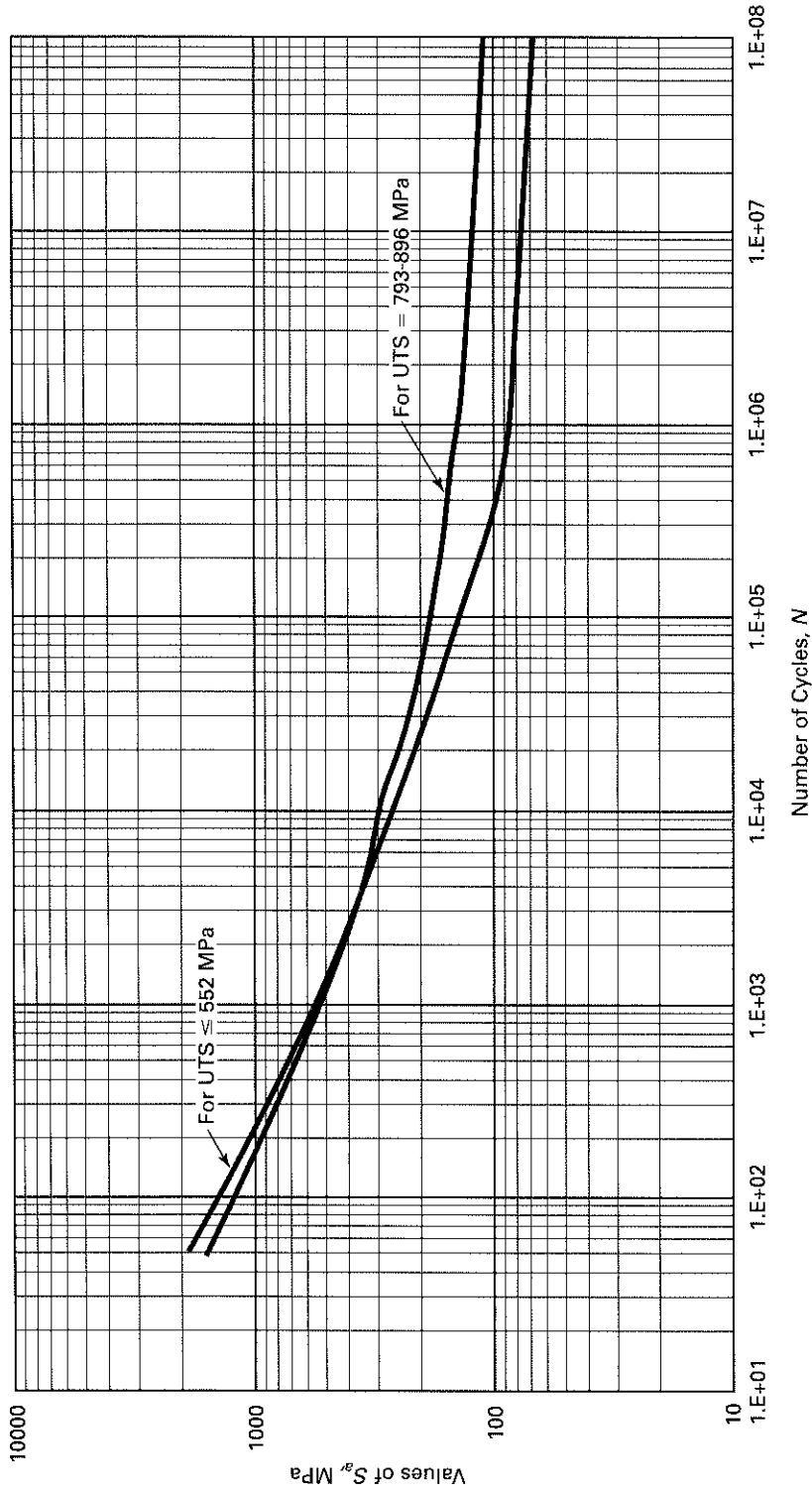
Figure KD-320.2
Design Fatigue Curve $S_a = f(N_f)$ for Nonwelded Parts Made of Carbon or Low Alloy Steels for Temperatures Not Exceeding 700°F and UTS Less Than 90 ksi



GENERAL NOTES:

- (a) $E = 30 \times 10^6$ psi.
- (b) Interpolate for UTS = 80–115 ksi.
- (c) Table KD-320.1 contains tabulated values and equations for these curves, and a formula for an accurate interpolation of these curves.

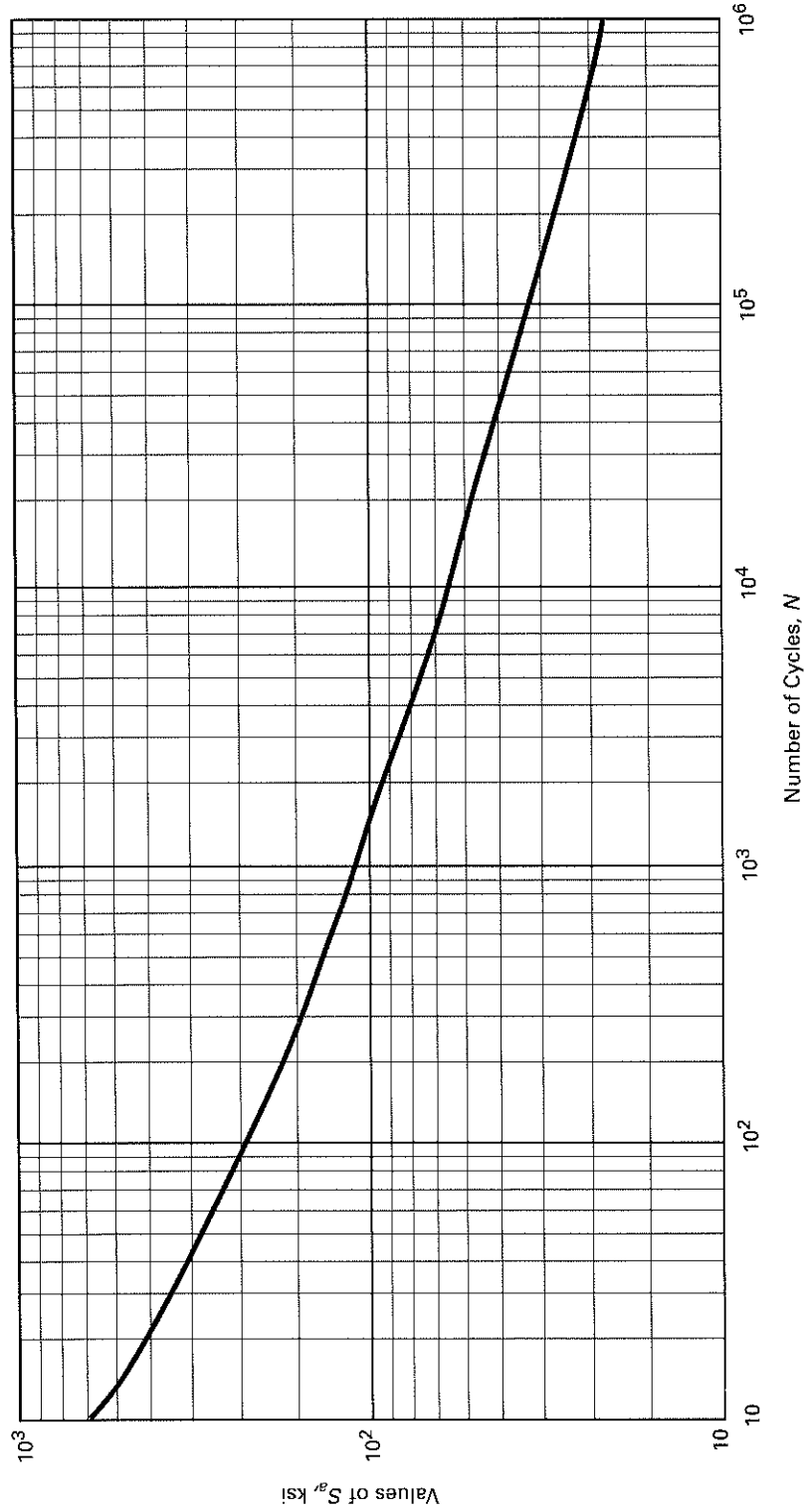
Figure KD-320.2M
Design Fatigue Curve $S_a = f(N_f)$ for Nonwelded Parts Made of Carbon or Low Alloy Steels for Temperatures Not Exceeding 371°C and UTS Less Than 620 MPa



GENERAL NOTES:

- (a) $E = 207 \times 10^3$ MPa.
- (b) Interpolate for $UTS = 552-793$ MPa.
- (c) Table KD-320.1M contains tabulated values and equations for these curves, and a formula for an accurate interpolation of these curves.

Figure KD-320.3
Design Fatigue Curve for Nonwelded Austenitic Stainless Steels for Temperatures Not Exceeding 800°F

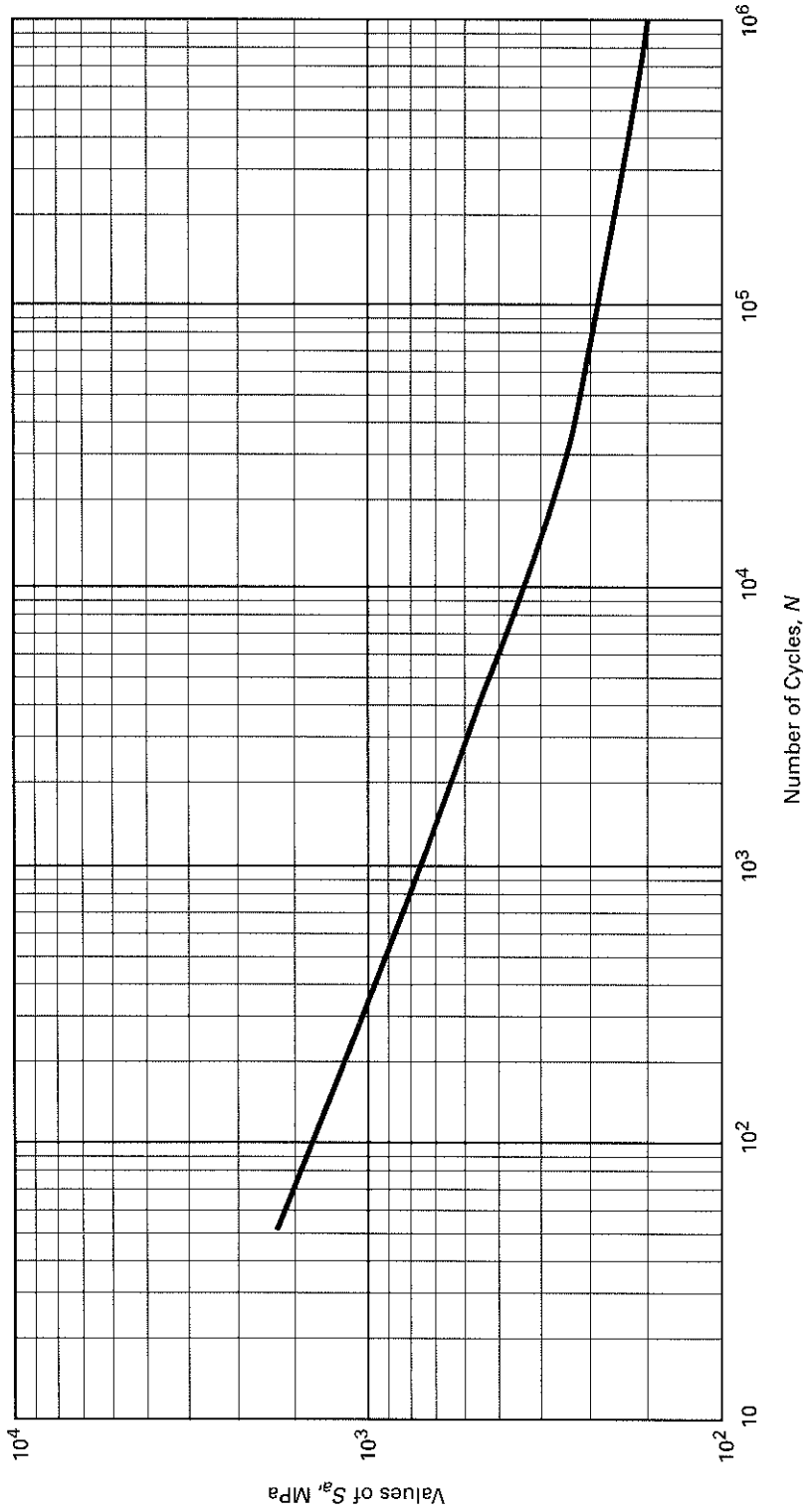


GENERAL NOTES:

(a) $E = 28.3 \times 10^6$ psi

(b) Table KD-320.1 contains tabulated values and an equation for this curve, and a equation for an accurate interpolation of this curve.

Figure KD-320.3M
Design Fatigue Curve for Nonwelded Austenitic Stainless Steels for Temperatures Not Exceeding 427°C

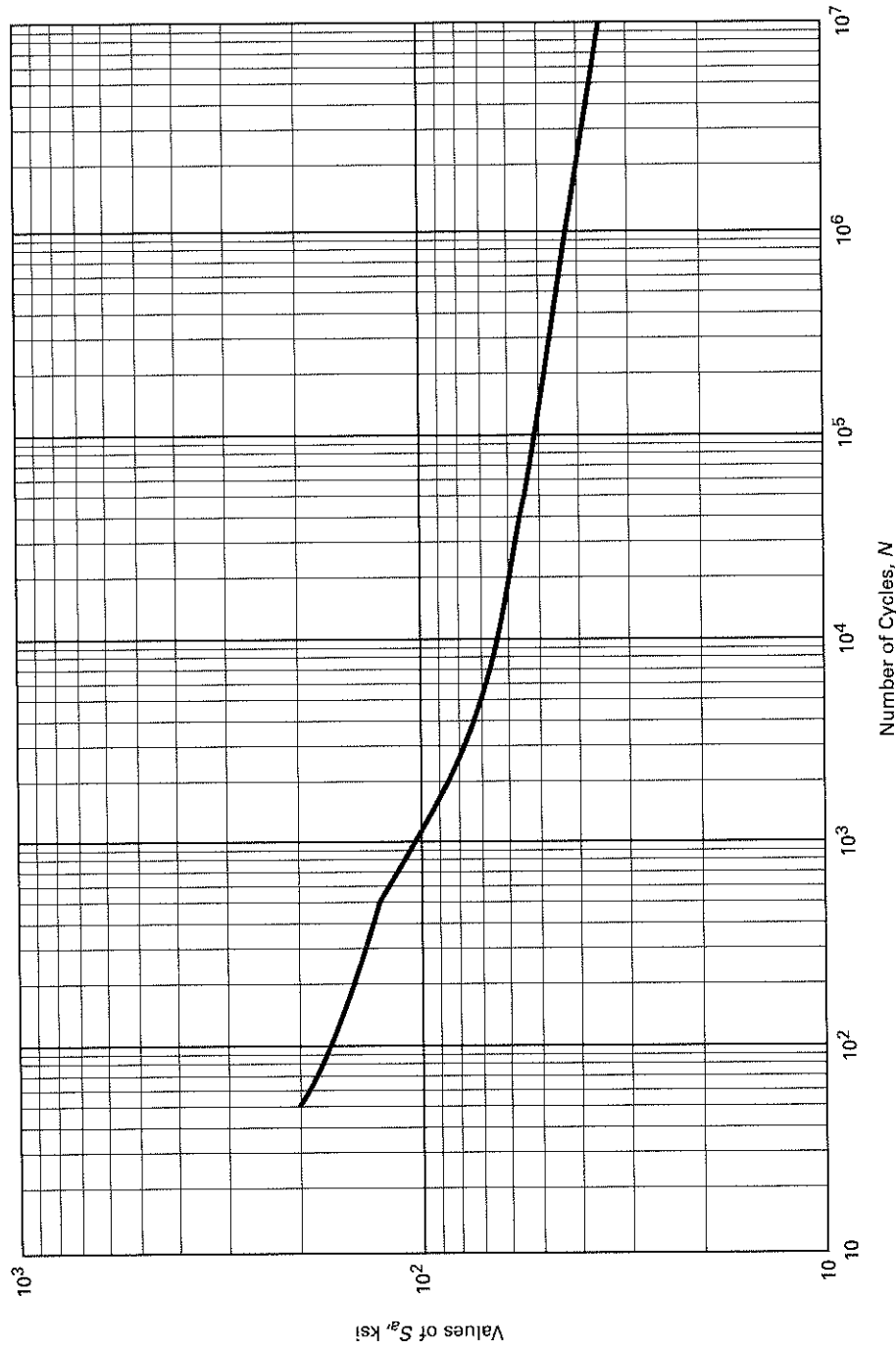


GENERAL NOTES:

(a) $E = 195 \times 10^3$ MPa

(b) Table KD-320.1M contains tabulated values and an equation for this curve, and a equation for an accurate interpolation of this curve.

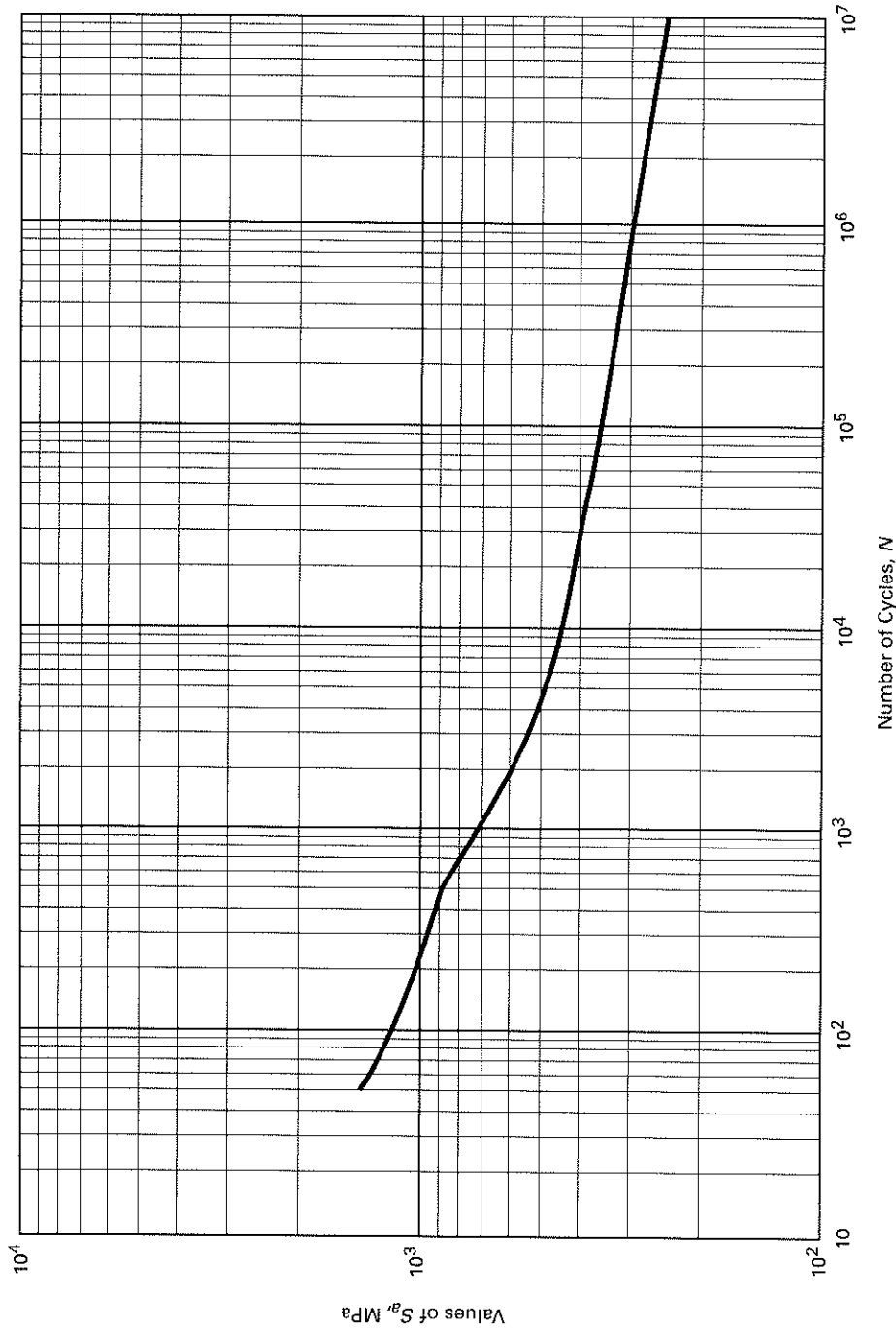
Figure KD-320.4
Design Fatigue Curve $S_a = f(N_f)$ for Nonwelded Machined Parts Made of 17-4PH/15-5PH Stainless Steel Bar or Forgings, for Temperatures Not Exceeding 550°F



GENERAL NOTES:

- (a) $E = 29.0 \times 10^6$ psi
- (b) Table KD-320.1 contains tabulated values and an equation for this curve, and a equation for the accurate interpolation of this curve.
- (c) When the 17-4PH/15-5PH heat treat condition is H1025 or stronger, this curve may be used only for analysis of either protective liners or inner layers in vessels demonstrated to be leak-before-burst (see KD-103 and KD-141).
- (d) Use of this curve is limited to cases where the peak Tresca strain range from mechanical loading is less than 1%.

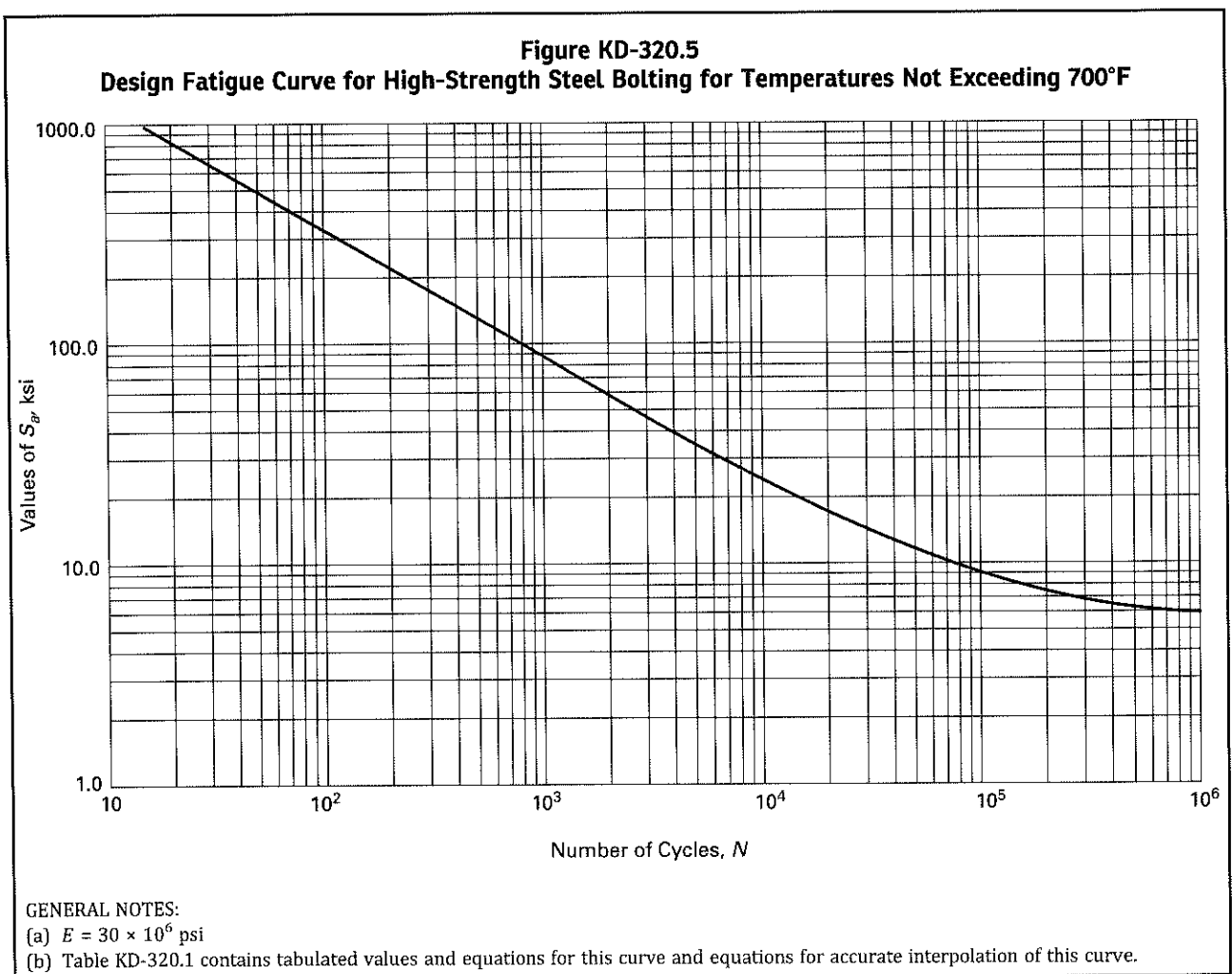
Figure KD-320.4M
Design Fatigue Curve $S_a = f(N_f)$ for Nonwelded Machined Parts Made of 17-4PH/15-5PH Stainless Steel Bar or Forgings, for Temperatures Not Exceeding 290°C



GENERAL NOTES:

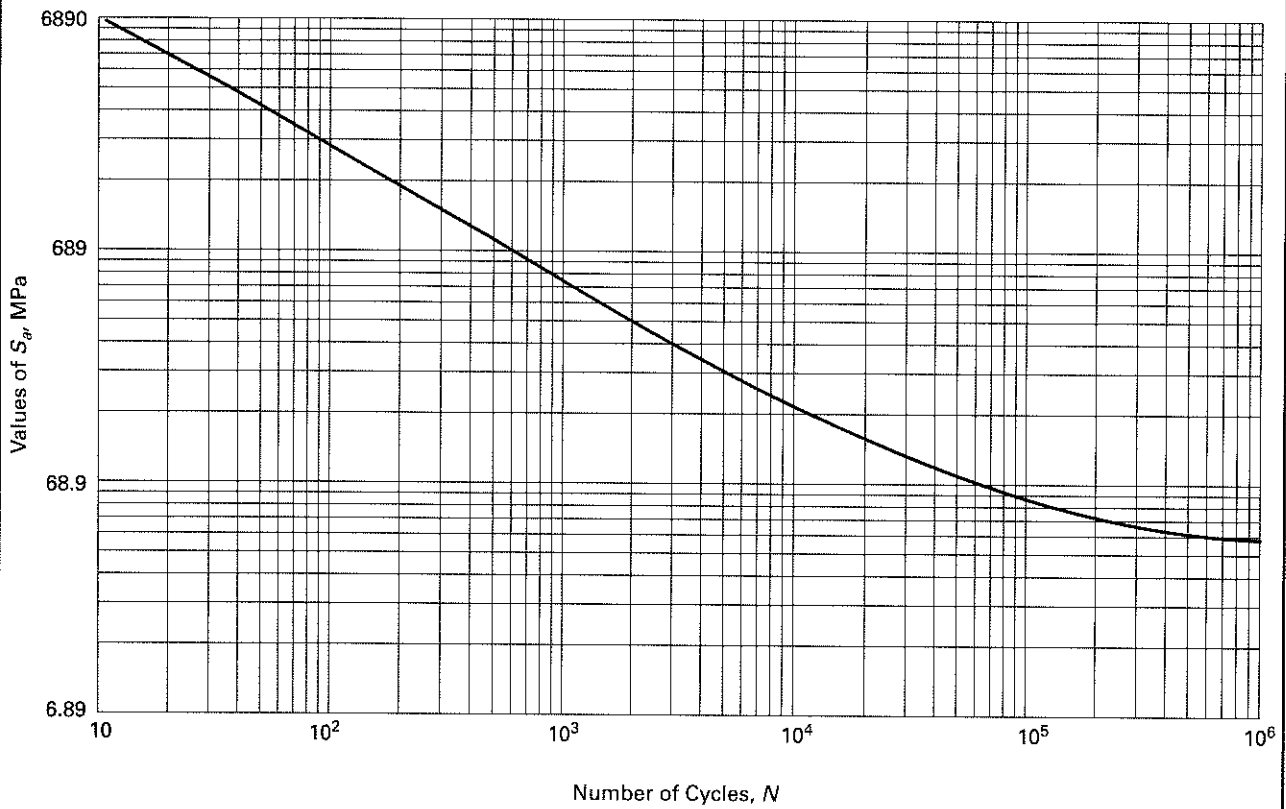
- (a) $E = 200 \times 10^3$ MPa
- (b) Table KD-320.1 contains tabulated values and an equation for this curve, and a equation for the accurate interpolation of this curve.
- (c) When the 17-4PH/15-5PH heat treat condition is H1025 or stronger, this curve may be used only for analysis of either protective liners or inner layers in vessels demonstrated to be leak-before-burst (see KD-103 and KD-141).
- (d) Use of this curve is limited to cases where the peak Tresca strain range from mechanical loading is less than 1%.

(13)



(13)

Figure KD-320.5M
Design Fatigue Curve for High-Strength Steel Bolting for Temperatures Not Exceeding 371°C

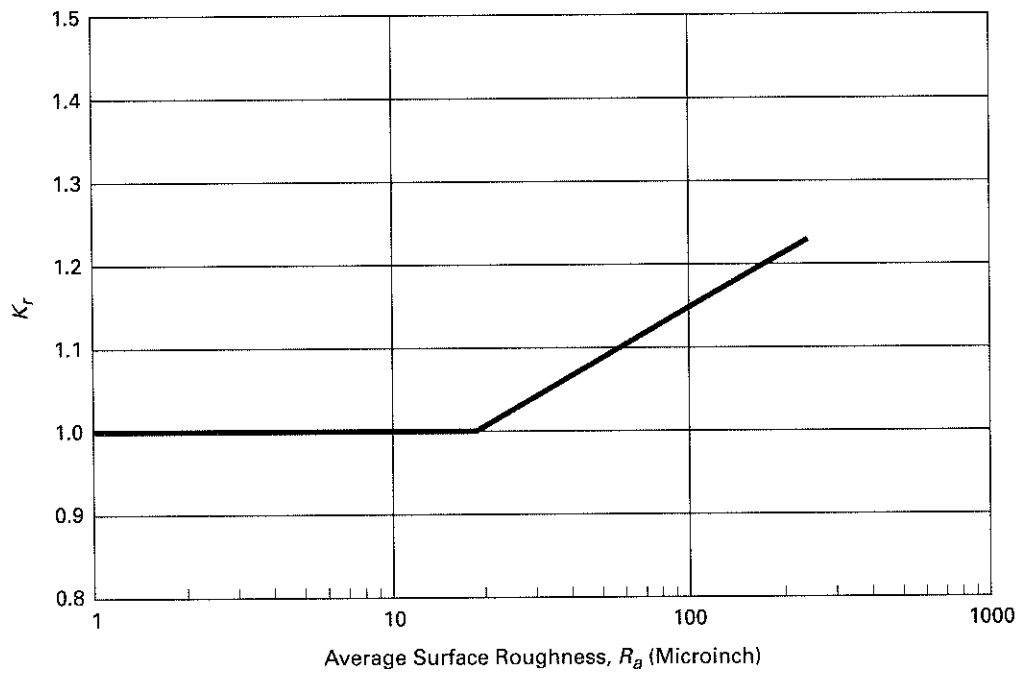


GENERAL NOTES:

- (a) $E = 206$ GPa
- (b) Table KD-320.1 contains tabulated values and equations for this curve and equations for accurate interpolation of this curve.

(13)

Figure KD-320.6(a)
Roughness Factor K_r Versus Average Surface Roughness R_a Microinch AA



GENERAL NOTES:

(a) See ASME B46.1 for definition of arithmetic average surface roughness, R_a .

(b) Curve equations:

$$R_a \leq 19 \text{ microinch}$$

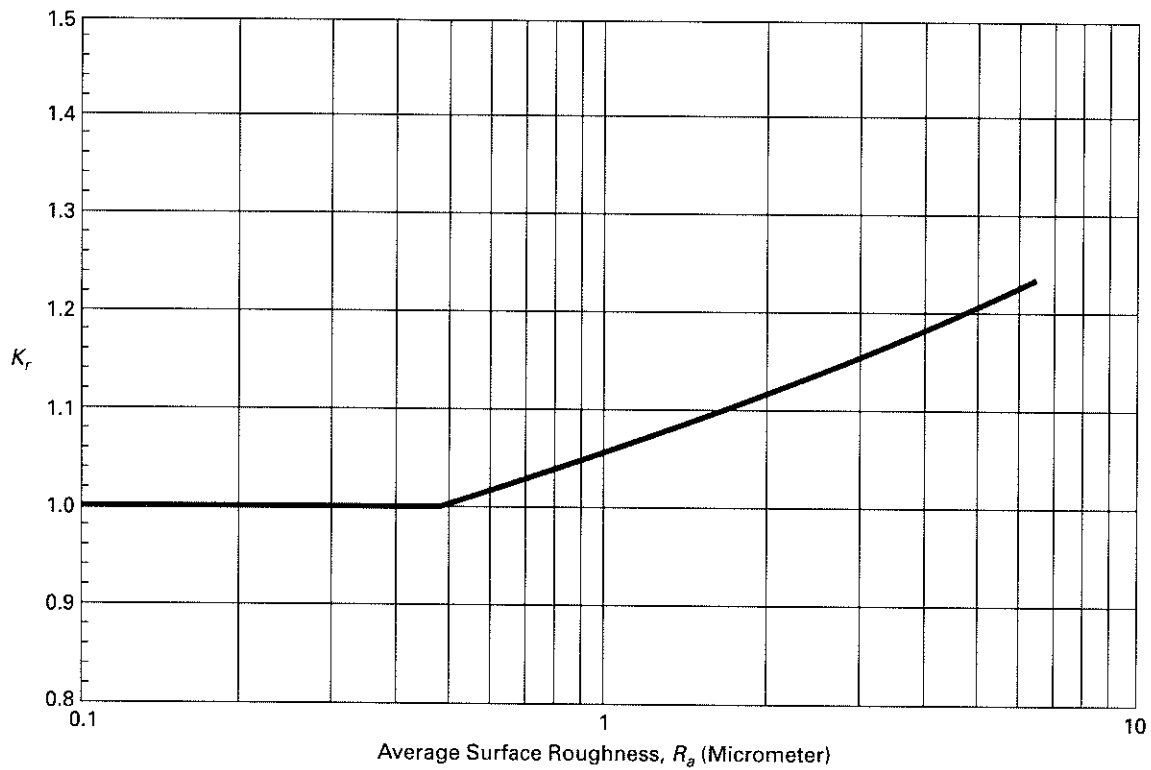
$$K_r = 1.0$$

$$19 < R_a \leq 250 \text{ microinch}$$

$$K_r = 1 / \{ -0.16998 \log [R_a \text{ (microinch)}] + 1.2166 \}$$

(13)

Figure KD-320.6M(a)
Roughness Factor K_r Versus Average Surface Roughness R_a Micrometer AA



GENERAL NOTES:

(a) See ASME B46.1 for definition of arithmetic average surface roughness, R_a .

(b) Curve equations:

$$R_a \leq 0.48 \text{ micrometer}$$

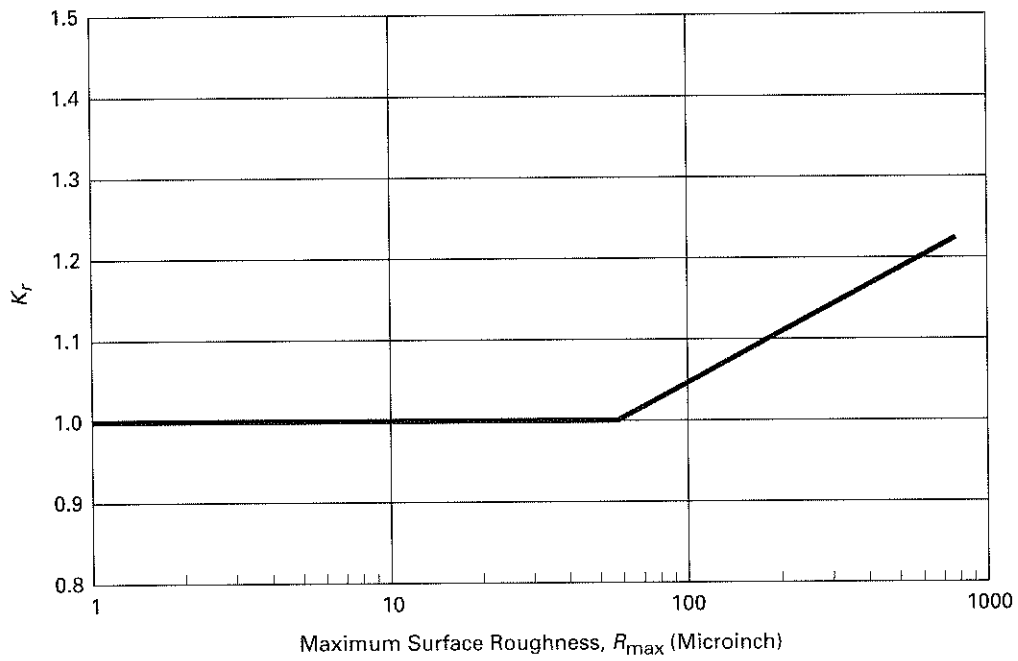
$$K_r = 1.0$$

$$0.5 < R_a \leq 6.4 \text{ micrometer}$$

$$K_r = 1 / \{ -0.16998 \log [R_a \text{ (micrometer)}] + 0.94545 \}$$

(13)

Figure KD-320.6(b)
Roughness Factor K_r Versus Maximum Surface Roughness R_{max} Microinch



GENERAL NOTES:

(a) See ASME B46.1 for definition of arithmetic average surface roughness, R_a .

(b) Curve equations:

$$R_{max} \leq 59 \text{ microinch}$$

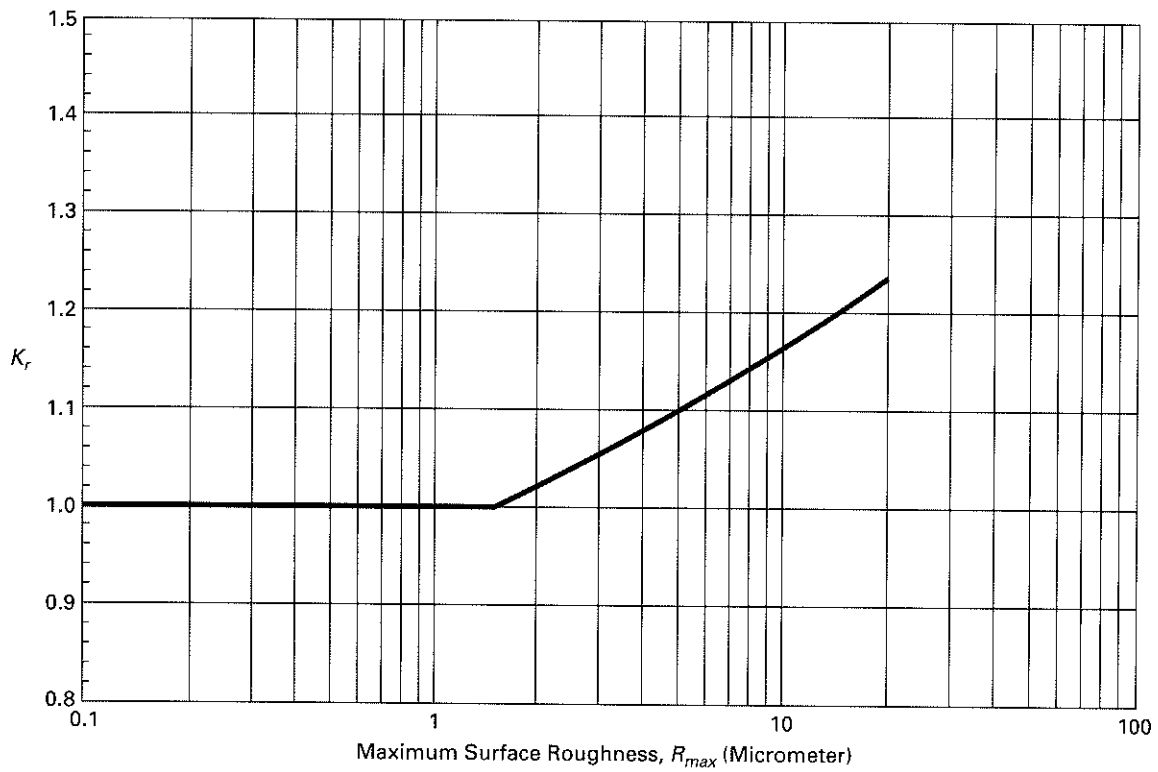
$$K_r = 1.0$$

$$59 < R_{max} \leq 785 \text{ microinch}$$

$$K_r = 1 / \{ -0.16998 \log [R_{max} (\text{microinch})] + 1.3011 \}$$

(13)

Figure KD-320.6M(b)
Roughness Factor K_r Versus Maximum Surface Roughness R_{max} Micrometer



GENERAL NOTES:

(a) See ASME B46.1 for definition of arithmetic average surface roughness, R_a .

(b) Curve equations:

$$R_{max} \leq 1.50 \text{ micrometer}$$

$$K_r = 1.0$$

$$1.50 < R_{max} \leq 20 \text{ micrometer}$$

$$K_r = 1 / \{ -0.16998 \log [R_{max} \text{ (micrometer)}] + 1.02995 \}$$

Table KD-320.7
Tabulated Values of S_a Alternating Stress Intensity from Figures KD-320.7 and KD-320.7M

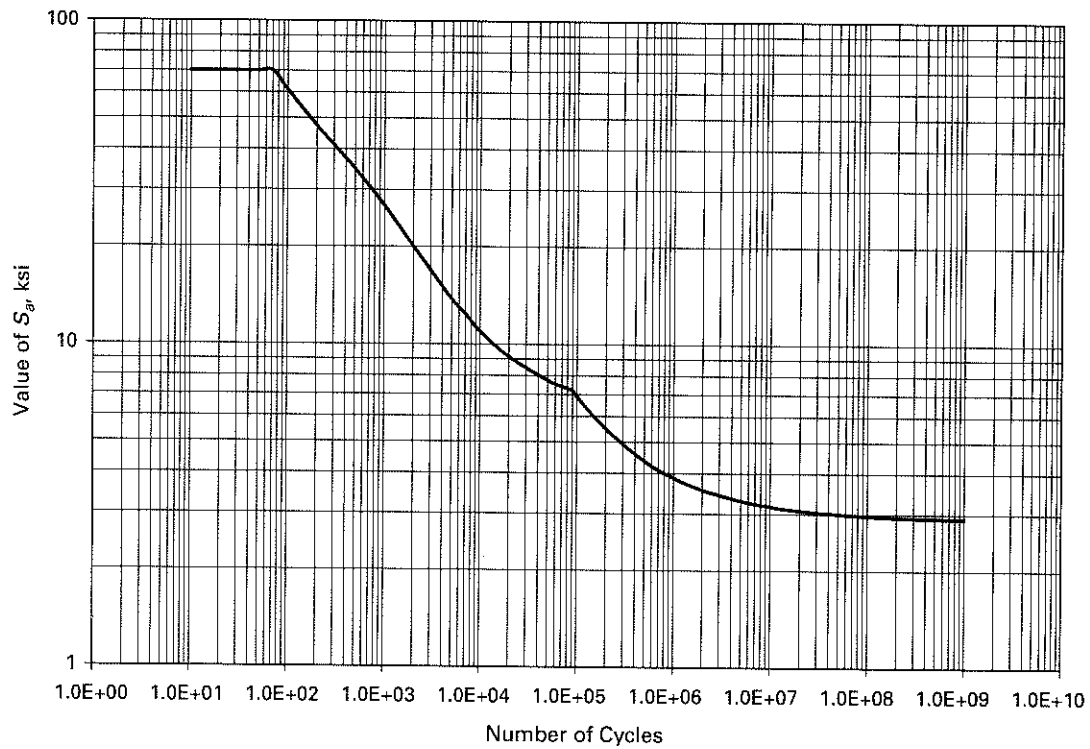
N, Number of Cycles	S_a With Maximum Mean Stress,	S_a With Maximum Mean Stress,
	ksi	MPa
1.0E+01	70.00	483
2.0E+01	70.00	483
5.0E+01	70.00	483
7.0E+01	70.00	483
1.0E+02	60.96	420
2.0E+02	47.20	325
5.0E+02	34.80	240
1.0E+03	26.79	185
2.0E+03	20.00	138
5.0E+03	13.78	95
7.0E+03	12.40	85
1.0E+04	10.93	75
2.0E+04	9.14	63
5.0E+04	7.74	53
9.0E+04	7.18	50
1.0E+05	6.89	48
2.0E+05	5.47	38
5.0E+05	4.36	30
1.0E+06	3.87	27
2.0E+06	3.55	24
5.0E+06	3.29	23
1.0E+07	3.16	22
2.0E+07	3.07	21
5.0E+07	3.00	21
1.0E+08	2.96	20
2.0E+08	2.93	20
5.0E+08	2.91	20
1.0E+09	2.90	20

GENERAL NOTES:

- (a) Number of design cycles indicated shall be as follows: $1.0E + J = 1 \times 10^J$,
 e.g., $5.0E + 2 = 5 \times 10^2$
- (b) Interpolation between tabular values is permissible based upon data representation by straight lines on a log-log plot.
 Accordingly, for $S_i > S > S_j$,
 $(N/N_i) = (N_j/N_i)^{[\log(S_i/S)/\log(S_i/S_j)]}$

(13)

Figure KD-320.7
Design Fatigue Curve for Nonwelded 6061-T6 and 6061-T651 Aluminum for Temperatures Not Exceeding 225°F

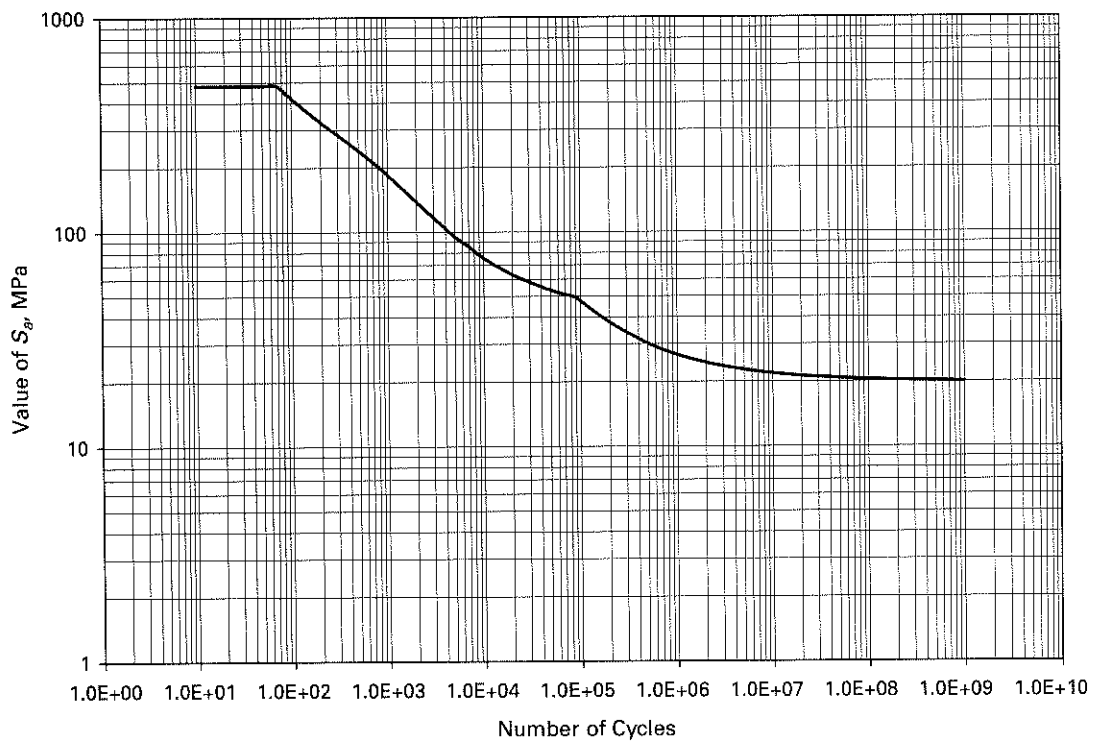


GENERAL NOTES:

(a) $E = 10 \times 10^6$ psi.

(b) Table KD-320.7 contains tabulated values.

Figure KD-320.7M
Design Fatigue Curve for Nonwelded 6061-T6 and 6061-T651 Aluminum for Temperatures Not Exceeding 107°C



GENERAL NOTES:

(a) $E = 69 \times 10^3$ MPa.

(b) Table KD-320.7 contains tabulated values.

Table KD-322.1
Fatigue Penalty Parameters

Material	m	n
Low alloy steel	2.0	0.2
Martensitic stainless steel	2.0	0.2
Carbon steel	3.0	0.2
Austenitic stainless steel	1.7	0.3
Nickel-chromium-iron	1.7	0.3
Nickel-copper	1.7	0.3

GENERAL NOTE: The fatigue penalty factor shall only be used if the component is not subjected to thermal ratcheting.

Table KD-360.1
Cyclic Stress-Strain Curve Data

Material Description	Temperature (°F)	n_{css}	K_{css} (ksi)
Carbon Steel (0.75 in.-base metal)	70	0.128	109.8
	390	0.134	105.6
	570	0.093	107.5
	750	0.109	96.6
Carbon Steel (0.75 in.-weld metal)	70	0.110	100.8
	390	0.118	99.6
	570	0.066	100.8
	750	0.067	79.6
Carbon Steel (2 in.-base metal)	70	0.126	100.5
	390	0.113	92.2
	570	0.082	107.5
	750	0.101	93.3
Carbon Steel (4 in.-base metal)	70	0.137	111.0
	390	0.156	115.7
	570	0.100	108.5
	750	0.112	96.9
1Cr- $\frac{1}{2}$ Mo (0.75 in.-base metal)	70	0.116	95.7
	390	0.126	95.1
	570	0.094	90.4
	750	0.087	90.8
1Cr- $\frac{1}{2}$ Mo (0.75 in.-weld metal)	70	0.088	96.9
	390	0.114	102.7
	570	0.085	99.1
	750	0.076	86.9
1Cr- $\frac{1}{2}$ Mo (2 in.-base metal)	70	0.105	92.5
	390	0.133	99.2
	570	0.086	88.0
	750	0.079	83.7
1Cr-1Mo- $\frac{1}{4}$ V	70	0.128	156.9
	750	0.128	132.3
	930	0.143	118.2
	1020	0.133	100.5
	1110	0.153	80.6
2 $\frac{1}{4}$ Cr- $\frac{1}{2}$ Mo	70	0.100	115.5
	570	0.109	107.5
	750	0.096	105.9
	930	0.105	94.6
	1110	0.082	62.1
9Cr-1Mo	70	0.177	141.4
	930	0.132	100.5
	1020	0.142	88.3
	1110	0.121	64.3
	1200	0.125	49.7
Type 304	70	0.171	178.0
	750	0.095	85.6
	930	0.085	79.8
	1110	0.090	65.3
	1290	0.094	44.4

Table KD-360.1
Cyclic Stress-Strain Curve Data (Cont'd)

Material Description	Temperature (°F)	n_{css}	K_{css} (ksi)
Type 304 (Annealed)	70	0.334	330.0
800H	70	0.070	91.5
	930	0.085	110.5
	1110	0.088	105.7
	1290	0.092	80.2
	1470	0.080	45.7
Aluminum (Al-4.5Zn-0.6Mn)	70	0.058	65.7
Aluminum (Al-4.5Zn-1.5Mg)	70	0.047	74.1
Aluminum (1100-T6)	70	0.144	22.3
Aluminum (2014-T6)	70	0.132	139.7
Aluminum (5086)	70	0.139	96.0
Aluminum (6009-T4)	70	0.124	83.7
Aluminum (6009-T6)	70	0.128	91.8
Copper	70	0.263	99.1

Table KD-360.1M
Cyclic Stress-Strain Curve Data

Material Description	Temperature °C	n_{css}	K_{css} (MPa)
Carbon Steel (20 mm-base metal)	20	0.128	757
	200	0.134	728
	300	0.093	741
	400	0.109	666
Carbon Steel (20 mm-weld metal)	20	0.110	695
	200	0.118	687
	300	0.066	695
	400	0.067	549
Carbon Steel (50 mm-base metal)	20	0.126	693
	200	0.113	636
	300	0.082	741
	400	0.101	643
Carbon Steel (100 mm-base metal)	20	0.137	765
	200	0.156	798
	300	0.100	748
	400	0.112	668
1Cr- $\frac{1}{2}$ Mo (20 mm-base metal)	20	0.116	660
	200	0.126	656
	300	0.094	623
	400	0.087	626
1Cr- $\frac{1}{2}$ Mo (20 mm-weld metal)	20	0.088	668

Table KD-360.1M
Cyclic Stress-Strain Curve Data (Cont'd)

Material Description	Temperature °C	n_{CSS}	K_{CSS} (MPa)
	200	0.114	708
	300	0.085	683
	400	0.076	599
1Cr- $\frac{1}{2}$ Mo (50 mm-base metal)	20	0.105	638
	200	0.133	684
	300	0.086	607
	400	0.079	577
1Cr-1Mo- $\frac{1}{4}$ V	20	0.128	1082
	400	0.128	912
	500	0.143	815
	550	0.133	693
	600	0.153	556
2- $\frac{1}{4}$ Cr- $\frac{1}{2}$ Mo	20	0.100	796
	300	0.109	741
	400	0.096	730
	500	0.105	652
	600	0.082	428
9Cr-1Mo	20	0.117	975
	500	0.132	693
	550	0.142	609
	600	0.121	443
	650	0.125	343
Type 304	20	0.171	1227
	400	0.095	590
	500	0.085	550
	600	0.090	450
	700	0.094	306
Type 304 (Annealed)	20	0.334	2275
800H	20	0.070	631
	500	0.085	762
	600	0.088	729
	700	0.092	553
	800	0.080	315
Aluminum (Al-4.5Zn-0.6Mn)	20	0.058	453
Aluminum (Al-4.5Zn-1.5Mg)	20	0.047	511
Aluminum (1100-T6)	20	0.144	154
Aluminum (2014-T6)	20	0.132	963
Aluminum (5086)	20	0.139	662
Aluminum (6009-T4)	20	0.124	577
Aluminum (6009-T6)	20	0.128	633
Copper	20	0.263	683

Table KD-370.1
Coefficients for the Welded Joint Fatigue Curves

Statistical Basis	All Alloys in KD-371 Except Aluminum		Aluminum Alloys	
	C	h	C	h
Mean Curve	1408.7	0.31950	247.04	0.27712
Upper 68% Prediction Interval (+1 σ)	1688.3	0.31950	303.45	0.27712
Lower 68% Prediction Interval (-1 σ)	1175.4	0.31950	201.12	0.27712
Upper 95% Prediction Interval (+2 σ)	2023.4	0.31950	372.73	0.27712
Lower 95% Prediction Interval (-2 σ)	980.8	0.31950	163.73	0.27712
Upper 99% Prediction Interval (+3 σ)	2424.9	0.31950	457.84	0.27712
Lower 99% Prediction Interval (-3 σ)	818.3	0.31950	133.29	0.27712

GENERAL NOTE: In U.S. Customary Units, the equivalent structural stress range parameter, ΔS_{range} , in L-320 and the structural stress effective thickness, t_{ess} , defined in KD-340 are in ksi/(inches)^{(2- m_{ss})/2 m_{ss}} and inches, respectively. The parameter m_{ss} is defined in KD-340.

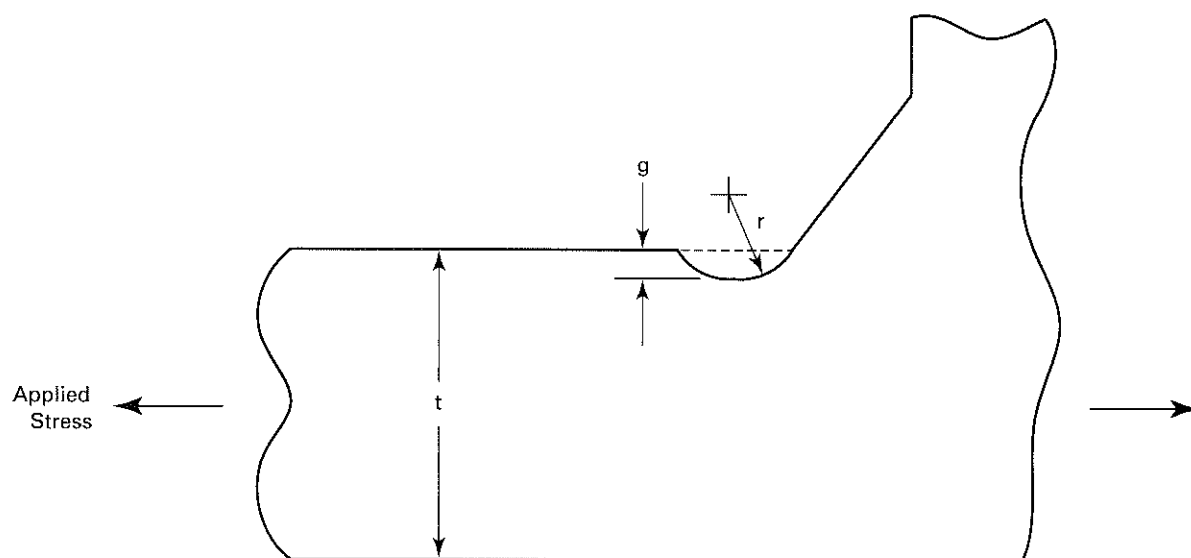
Table KD-370.1M
Coefficients for the Welded Joint Fatigue Curves

Statistical Basis	All Alloys in KD-371 Except Aluminum		Aluminum Alloys	
	C	h	C	h
Mean Curve	19930.2	0.31950	3495.13	0.27712
Upper 68% Prediction Interval (+1 σ)	23885.8	0.31950	4293.19	0.27712
Lower 68% Prediction Interval (-1 σ)	16629.7	0.31950	2845.42	0.27712
Upper 95% Prediction Interval (+2 σ)	28626.5	0.31950	5273.48	0.27712
Lower 95% Prediction Interval (-2 σ)	13875.7	0.31950	2316.48	0.27712
Upper 99% Prediction Interval (+3 σ)	34308.1	0.31950	6477.60	0.27712
Lower 99% Prediction Interval (-3 σ)	11577.9	0.31950	1885.87	0.27712

GENERAL NOTE: In SI Units, the equivalent structural stress range parameter, ΔS_{range} , in L-320 and the structural stress effective thickness, t_{ess} , defined in KD-340 are in MPa/(mm)^{(2- m_{ss})/2 m_{ss}} and mm, respectively. The parameter m_{ss} is defined in KD-340.

(13)

**Figure KD-372.1
Burr Grinding of Weld Toe**



$g = 0.5 \text{ mm (0.02 in.) below undercut;}$
 $r \geq 0.25t \geq 4g$

ARTICLE KD-4

FRACTURE MECHANICS EVALUATION

KD-400 SCOPE

This Article presents a fracture mechanics design approach. In accordance with KD-140, if it can be shown that the vessel will fail in a leak-before-burst mode, then the number of design cycles shall be calculated in accordance with either Article KD-3 or Article KD-4. If a leak-before-burst mode of failure cannot be shown, then the number of design cycles shall be calculated in accordance with this Article.

(13) KD-401 GENERAL

(a) This Article is based on the assumption that the crack initiation stage is complete and that cracks exist at highly stressed points in the pressure vessel. Cracks are defined in this Division as flaws that are predominantly characterized by length and depth, with a sharp root radius. The principles of linear elastic fracture mechanics were used to develop the criteria in this Article for calculating the number of design cycles to propagate these cracks to the critical crack depth and the maximum allowable depth. See also Nonmandatory Appendix D.

(b) Manufacturing processes such as welding, heat treatment, forming, autofrettage, shrink fitting, and wire wrapping introduce residual stresses. Some cracks may propagate through the resulting residual stress field due to cyclic loading. A method for accounting for these residual stresses is given in KD-420.

(c) The critical crack depth for a given loading condition is defined to the crack depth that is calculated using the failure assessment diagram from API 579-1/ASME FFS-1. The critical crack depth shall be calculated for the most severe combination of loading conditions. If the critical crack depth is less than the wall thickness, it may not be possible to assume a leak-before-burst mode of failure. However, see KD-141.

(d) For aluminum alloys listed in Table KM-400-4 or Table KM-400-4M, a value of $23 \text{ ksi-in.}^{1/2}$ ($25 \text{ MPa-m}^{1/2}$) shall be used for the critical stress intensity factor, K_{IC} , in the fracture mechanics evaluation.

KD-410 CRACK SIZE CRITERIA

KD-411 ASSUMED INITIAL CRACK SIZE

(a) The initial crack size to be used for the calculation of the crack propagation design cycles shall be based on the nondestructive examination method to be used. If the nondestructive examination method to be used measures the

length of a surface indication, an assumption shall be made as to the depth and shape of an assumed initial crack.

(b) A surface crack not associated with a stress concentration shall be assumed to be semielliptical with a ratio of depth to surface length of 1:3. The assumed surface length shall not be less than the maximum acceptable nondestructive examination indication as given in Part KE unless a smaller length is specified in the User's Design Specification [see KG-311.12(a)]. If a smaller length is specified, it must be clearly demonstrated that the nondestructive examination method used will reliably detect indications of that size.

(c) For a thread root or circumferential groove, the crack shall be assumed to be annular. The initial crack depth shall be assumed to be not less than one-third of the maximum acceptable length of a surface nondestructive examination indication as defined above.

KD-412 ALLOWABLE FINAL CRACK DEPTH

To calculate the number of design cycles N_p based on crack propagation, it is necessary to determine an allowable final crack depth. The allowable final crack depth shall be calculated using the failure assessment diagram (FAD) in API 579-1/ASME FFS-1. The calculated number of design cycles is the number of cycles required to propagate a crack of the assumed initial flaw size to that allowable final crack depth. The calculated number of design cycles is defined as the lesser of

(a) the number of cycles corresponding to one-half of the number of cycles required to propagate a crack from the initial assumed flaw size to the critical crack depth [see KD-401(c)]

(b) the number of cycles required to propagate a crack from the initial assumed flaw size to the depth as defined in KD-412.1 and KD-412.2

KD-412.1 Monobloc Vessels. For monobloc vessels, the allowable final crack depth shall be the lesser of 25% of the section thickness being considered or 25% of the critical crack depth.

KD-412.2 Vessels With Two or More Layers.

(a) For vessels with two or more layers, the final crack depth in the inner layer may be equal to the layer thickness, provided the theoretical collapse pressure (1.732 times the value calculated in KD-221.2) of the combined remaining layers is at least 20% higher than the design pressure of the unflawed vessel.